

Software Requirements Specification

Course: CSE 3411 – SAD Lab

Project Title: Nexus Portal

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of Nexus Portal, a unified opportunity management system designed for students, faculty members, employers, and administrators. This document serves as a comprehensive reference for understanding the system's objectives, features, constraints, and expected behavior.

The SRS provides a clear description of what the system will do, the problems it aims to solve, and the requirements that guide its design and development. It is intended to ensure a shared understanding among all stakeholders involved in the development and evaluation of the system.

1.2 Intended Audience

This document is intended for multiple stakeholders involved in the Nexus Portal project, including:

- Course instructors and evaluators for assessment and review purposes.
- Project team members responsible for system analysis, design, and implementation.
- Future developers who may extend or maintain the system.
- Academic administrators and decision-makers interested in understanding the system's capabilities and feasibility.

The SRS should be used as a baseline reference throughout the system's lifecycle, including design validation, development planning, and future enhancements.

1.3 Scope of the System

Nexus Portal is designed to provide a centralized digital platform that integrates job opportunities, internships, and research assistantships into a single, verified ecosystem.

The system's primary goal is to eliminate opportunity fragmentation, improve trust through verification mechanisms, and streamline the recruitment and application process for all stakeholders.

The system supports the following major functions:

- A unified opportunity feed for jobs, internships, and research positions.
- Role-based access for students, faculty members, employers, and administrators.
- University-based user verification to reduce fake postings and resume fraud.
- Application tracking with real-time status updates.
- Resume generation and management tools for students.
- Research collaboration features for faculty and research assistants.
- Administrative moderation, verification, and reporting tools.

The scope of Nexus Portal is limited to the web-based and responsive system design required for the academic project. External integrations such as direct university database access and advanced AI-based features are considered future enhancements and are outside the scope of the current implementation.

2. System Overview

2.1 Product Perspective

The NexusPortal is a web-based, centralized career and research management system designed to integrate job opportunities, internships, and academic research positions and management research tasks into a single platform.

NexusPortal operates as an independent system but interacts with external services such as:

- Email/SMS gateways for verification and notifications
- Cloud storage for resume and document uploads

- Calendar services for interview scheduling (optional integration)

Unlike traditional job portals that focus only on employment, NexusPortal extends functionality to academic research collaboration, enabling faculty members to recruit students for research assistantships and lab work.

The system follows a client-server architecture, where:

- The frontend acts as the client interface for users
- The backend manages business logic, authentication, and workflows
- A centralized database stores user profiles, opportunities, and applications

2.2 Product Functions

At a high level, the NexusPortal platform provides the following core functions:

- ❖ User registration and role-based authentication {Student(Employee), Employer, Faculty, Admin}
- ❖ Verified posting of:
 - Job opportunities
 - Internship opportunities
 - Research assistantship and lab opportunities
- ❖ Advanced search and filtering of opportunities based on skills, domain, location, and eligibility
- ❖ Student profile management including skills, education, projects, and research interests
- ❖ Integrated resume builder with multiple templates and version control
- ❖ One-click application and application status tracking

- ❖ Research collaboration features such as research group creation and student participation
- ❖ Messaging and interview scheduling between applicants and recruiters
- ❖ Notification and deadline reminder system
- ❖ Feedback and rating mechanism to establish trust
- ❖ Administrative monitoring, verification, analytics, and reporting

2.3 User Classes and Characteristics

The system supports multiple user classes with distinct responsibilities and access levels.

2.3.1 Student / Job Seeker

- Undergraduate or graduate students and fresh graduates
- Limited professional experience
- Requires guidance, resume support, and opportunity tracking
- Access to job, internship, and research postings
- Can apply, track applications, and join research groups

2.3.2 Employer / Industry HR

- Company representatives or HR personnel
- Authorized to post job and internship opportunities
- Requires applicant filtering, resume review, and communication tools
- Interested in verified and qualified candidates

2.3.3 University Faculty / Research Supervisor

- University teachers, professors, or lab supervisors
- Can post research opportunities and create research groups
- Evaluates students based on skills, academic background, and contributions
- Tracks research progress and collaboration

2.3.4 Administrator

- System owner or career center authority
- Full system access
- Responsible for user verification, opportunity approval, dispute resolution, and analytics

2.3.5 Reviewer / Validator(Admin as a Special Role)

- Assigned staff or moderators
- Reviews postings for authenticity, employer's documents and policy compliance
- Prevents fake or misleading opportunities

2.4 Operating Environment

The Unified Opportunity Platform will operate in the following environment:

Hardware Requirements

- Server: Cloud-based or on-premise server with minimum 8GB RAM
- Client devices: Desktop, laptop, tablet, or smartphone

Software Requirements

- Operating System: Windows, Linux, or macOS
- Web Server: Apache or Nginx
- Backend Framework: Node.js / Django
- Database Management System: PostgreSQL / MySQL
- Frontend: HTML5, CSS3, JavaScript (React)
- Browser Support: Chrome, Firefox, Edge, Safari

Network Requirements

- Internet connectivity with standard HTTP/HTTPS support
- Secure communication using SSL/TLS

2.5 Design and Implementation Constraints

The system design and implementation are subject to the following constraints:

- Must follow role-based access control to ensure security.
- User data privacy regulations must be respected.
- Verification workflows may require manual admin intervention.
- The system must be scalable to support multiple universities and organizations.
- Development is constrained by limited time and resources typical of an academic project.
- Rule-based recommendation system only (no AI/ML implementation in initial phase)

2.6 Assumptions and Dependencies

Assumptions

- Users provide accurate and truthful information during registration.
- Employers and faculty cooperate in verification processes.
- Students have basic internet access and digital literacy.
- Universities support the use of institutional email for verification.

Dependencies

- Email and SMS services for OTP and notifications.
- Cloud storage for document uploads.
- Availability of administrators for manual review.
- Reliable hosting infrastructure.

2.7 Software Development Life Cycle (SDLC)

The development of the **NexusPortal** follows a **Waterfall Model with Iterative Enhancements**. This approach is suitable because the system requirements are well-defined at the beginning, while still allowing controlled improvements after each phase.

2.7.1 Requirement Analysis Phase

In this phase, the overall objectives and functional scope of the Unified Opportunity Platform are identified.

- Identification of stakeholders: students, employers, faculty members, and administrators.
- Analysis of problems in existing job, internship, and research opportunity systems.
- Collection of functional and non-functional requirements.
- Documentation of system requirements in the Software Requirements Specification (SRS).

Outcome: Clearly defined and approved system requirements.

2.7.2 System Design Phase

This phase focuses on translating requirements into a structured system design.

- Design of overall system architecture (client–server model).
- Definition of system modules and their interactions.
- Design of role-based user interfaces and dashboards.
- Security and access control design.

Outcome: Complete system design documentation.

2.7.3 Implementation Phase

During implementation, the system design will be converted into executable code.

- Development of frontend components for different user roles.
- Backend development for authentication, opportunity management, and application tracking.
- Database creation and integration.
- Implementation of verification, notification, and messaging modules.

Outcome: A working software system with core functionalities implemented.

2.7.4 Testing Phase

The testing phase ensures that the system functions correctly and meets the specified requirements.

- Unit testing of individual modules.

- Integration testing between frontend, backend, and database.
- Validation of security, role access, and verification workflows.
- User acceptance testing to confirm usability and correctness.

Outcome: A stable and reliable system free from critical defects.

2.7.5 Deployment Phase

After successful testing, the system is deployed to the target environment.

- Deployment on a cloud-based or institutional server.
- Configuration of databases and application servers.
- Enabling secure access using HTTPS.
- Making the system available to end users.

Outcome: The system becomes operational and accessible to users.

2.7.6 Maintenance and Enhancement Phase

This phase involves continuous monitoring and improvement of the system.

- Fixing bugs and resolving reported issues.
- Performance optimization.
- Adding new features such as advanced analytics or integrations.
- Updating security measures and verification policies.

Outcome: Improved system reliability, scalability, and user satisfaction.

3. Requirements Elicitation & Analysis

3.1 Benchmark Analysis

A benchmark analysis was conducted to evaluate existing platforms and identify market gaps within the student and research recruitment space.

Benchmark Products: LinkedIn, Bdjobs, ResearchGate, Upwork, Glassdoor.

Comparison Summary:

- **LinkedIn & Bdjobs:** These platforms focus on traditional job listings but lack dedicated, verified modules for academic research assistantships.
- **ResearchGate:** While effective for research networking, it does not provide integrated tools for undergraduate internship tracking or verified student skill badges.
- **Upwork:** Focuses on freelance contracts and includes a verification layer, but it is not tailored to the university academic ecosystem.
- **Glassdoor:** Primarily utilized for company reviews and salary transparency, but it does not support research recruitment or specialized student internship workflows.

Nexus Portal's Differentiation: Nexus Portal provides a unique, unified feed that combines jobs, internships, and research assistantships with a strict "University Verified" layer to eliminate fraud.

Feature Category	Job Listings	Internship Listings	Research / RAListings	Verification Layer	CV Builder	In-App Messaging	Application Tracking	Research Collaboration Module	Interview Scheduling	Ratings
Nexus Portal	Yes	Yes	Yes	Yes	yes	Yes	Yes	Yes	Yes	Yes
LinkedIn	Yes	Yes	No	No	Yes	Yes	Yes	No	No	Yes
Bdjobs	Yes	No	No	No	No	No	No	No	No	No
ResearchGate	Yes	No	Yes	No	No	Yes	No	No	No	No
Upwork	No	No	No	Yes	No	Yes	Yes	No	Yes	Yes
Glassdoor	Yes	No	No	No	No	No	No	No	No	Yes

3.2 Requirements Analysis

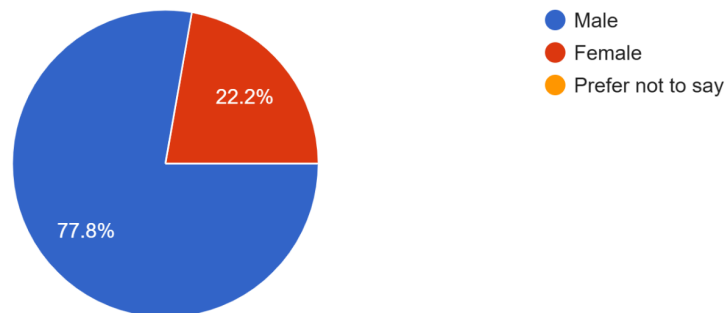
The requirement gathering process involved identifying internal project motivations and analyzing external market feedback through structured surveys. The purpose was to validate the challenges of opportunity fragmentation, gauge the demand for verified profiles, and prioritize high-value features like the research collaboration module.

3.2.1 Information Collected

- **Users:** A survey of 68 respondents was conducted, consisting of 95.6% students and job seekers. 50.8% of respondents rated application tracking as a significant "nightmare" pain level.
- **Faculty & Employers:** Faculty reported challenges with unqualified applicants and a lack of skill verification tools. Employers identified resume fraud and a lack of practical skills as their primary hiring hurdles.
- **Processes:** The system streamlines the process of finding opportunities across multiple platforms into one dashboard, moving from manual class announcements to digital research recruitment and verified application status tracking.

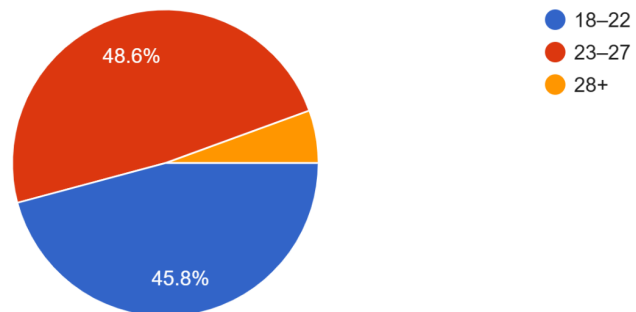
What is your Gender?

72 responses



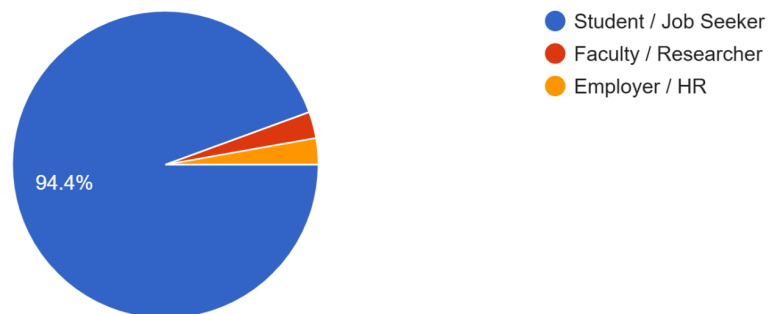
Which age group do you belong to?

72 responses



What is your current Occupation?

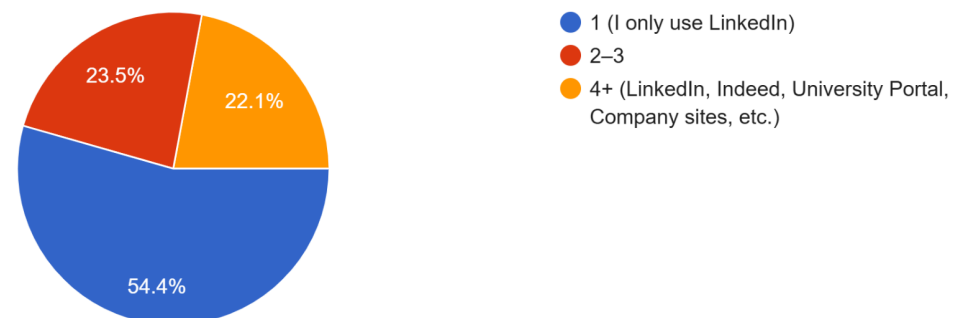
72 responses



Student Experience: Current Challenges in Opportunity Hunting

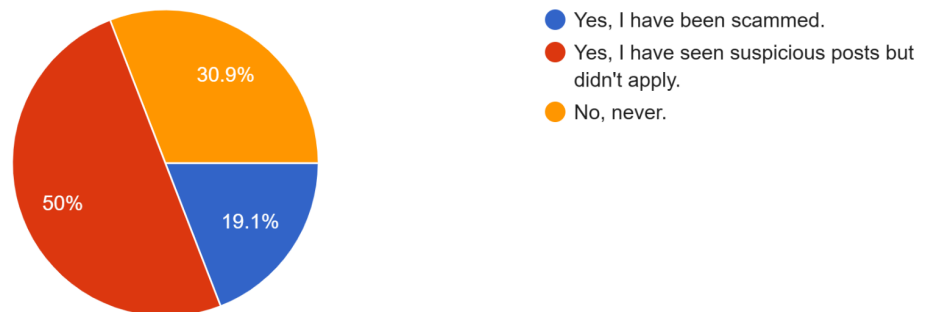
How many platforms do you currently check to find Jobs, Internships, and Research?

68 responses



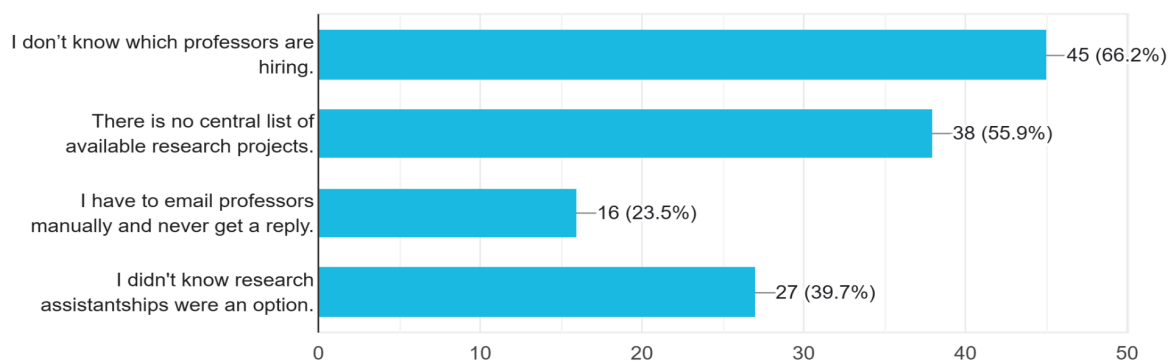
Have you ever encountered a "Fake" or "Ghost" job posting?

68 responses



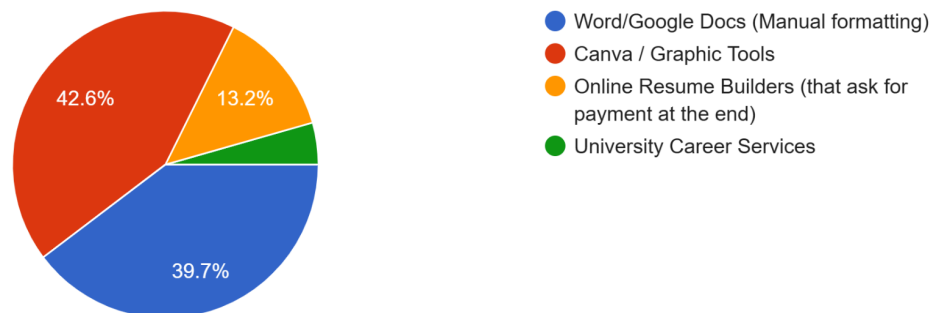
What is your biggest frustration with current University Research recruitment?

68 responses



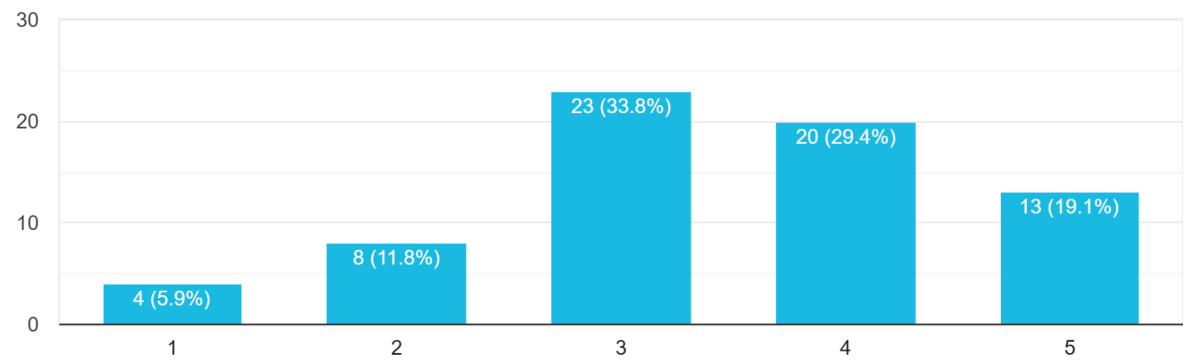
How do you currently create your resume?

68 responses



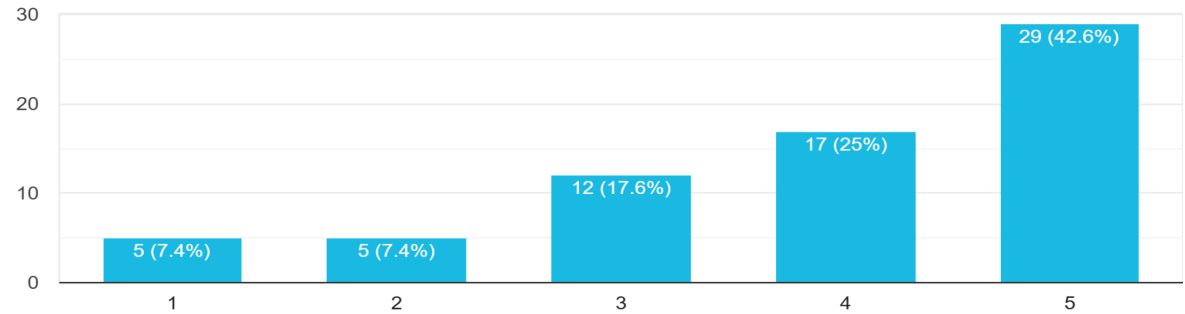
Rate your pain level on "Tracking Applications" (1 = Easy, 5 = Nightmare):

68 responses



If a platform existed that combined Internships AND Research positions in one feed, how likely would you use it? (

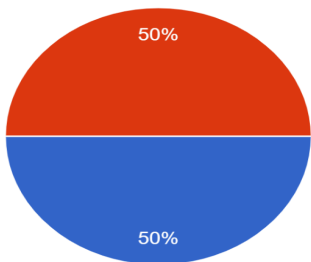
68 responses



Faculty Insight: Research Recruitment & Management

Would you find a "Weekly Student Contribution Tracker" useful for managing your research group?

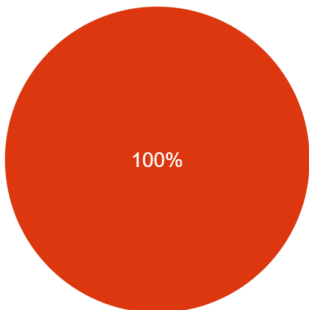
2 responses



- Yes, I desperately need this.
- Maybe, if it's simple to use.
- No, I prefer manual management.

What is the biggest challenge when hiring a student researcher?

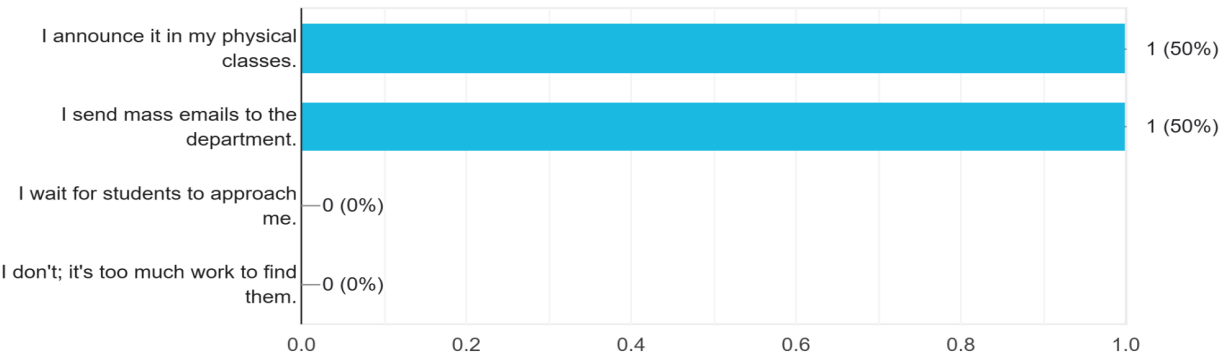
2 responses



- Too many unqualified applicants.
- Students lack specific technical skills (e.g., Python, Lab protocols).
- It takes too long to filter through emails/CVs.
- Students drop out of the research group mid-project.

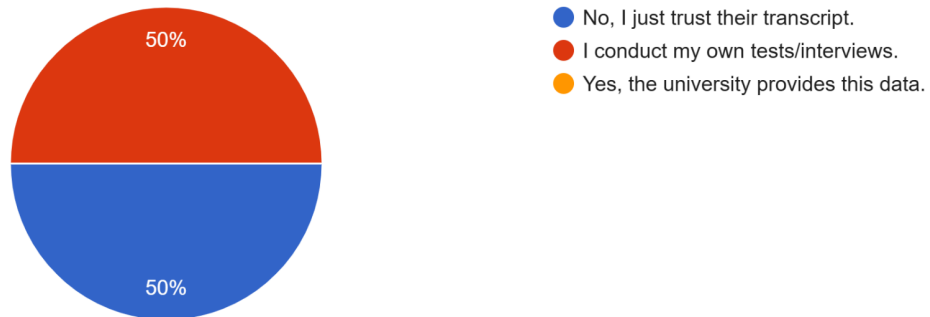
How do you currently recruit Research Assistants (RAs) for your projects? (Check all that apply)

2 responses



Do you currently have a way to verify a student's skills before they join your lab?

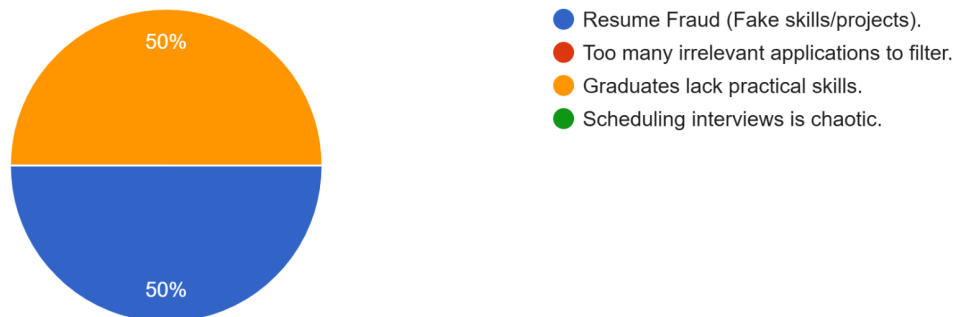
2 responses



Employer Perspective: Hiring Efficiency & Verification

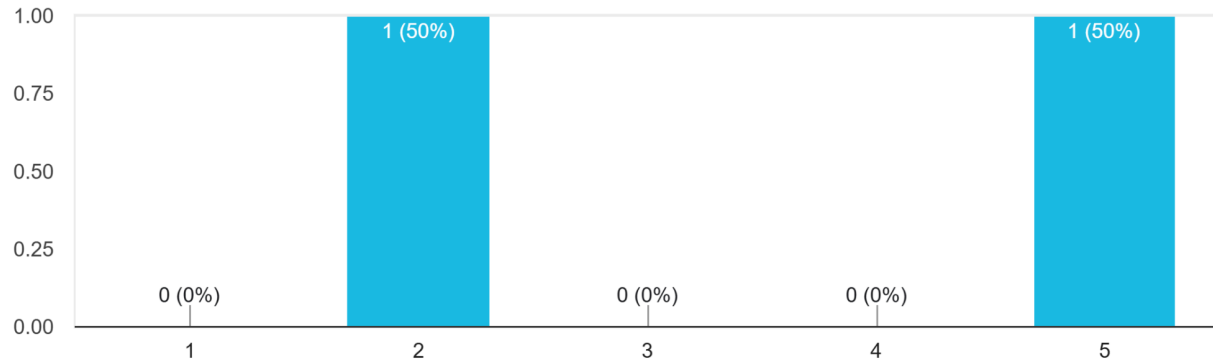
What is the hardest part of hiring fresh graduates?

2 responses



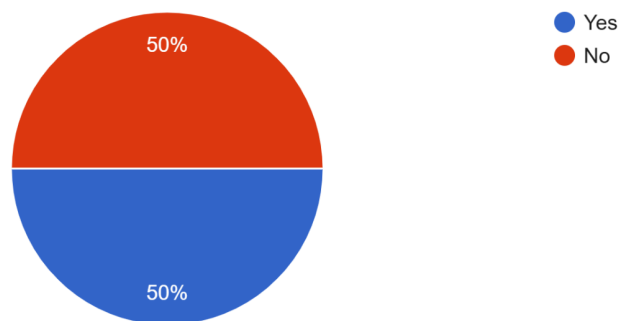
How important is "University Verified" status (ID/Transcript verification) when shortlisting candidates?

2 responses



Would you use a built-in "Interview Scheduler" that syncs with the applicant's status?

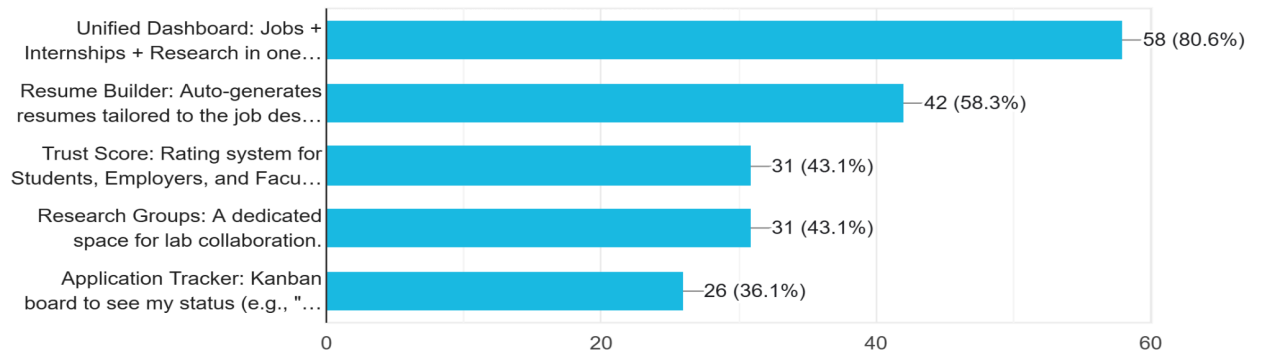
2 responses



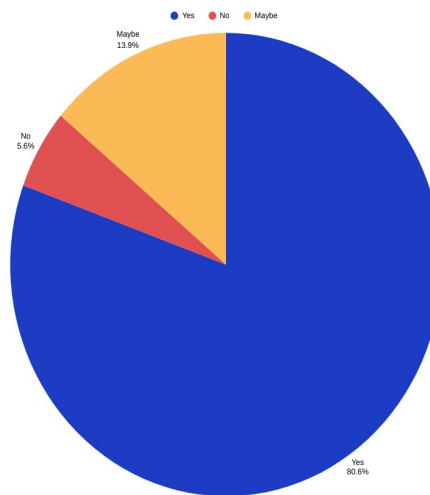
Proposed Solution: Feature Evaluation

We are building "NexusPortal" to solve these issues. Which feature excites you the most? (Check top 2)

72 responses

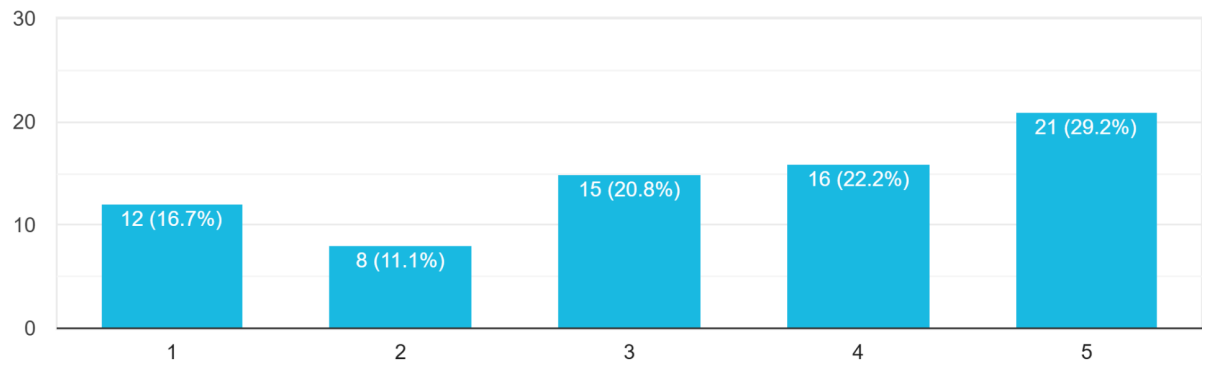


Would you trust a profile more if it had a "university verified" badge (ID/Transcript checked)?



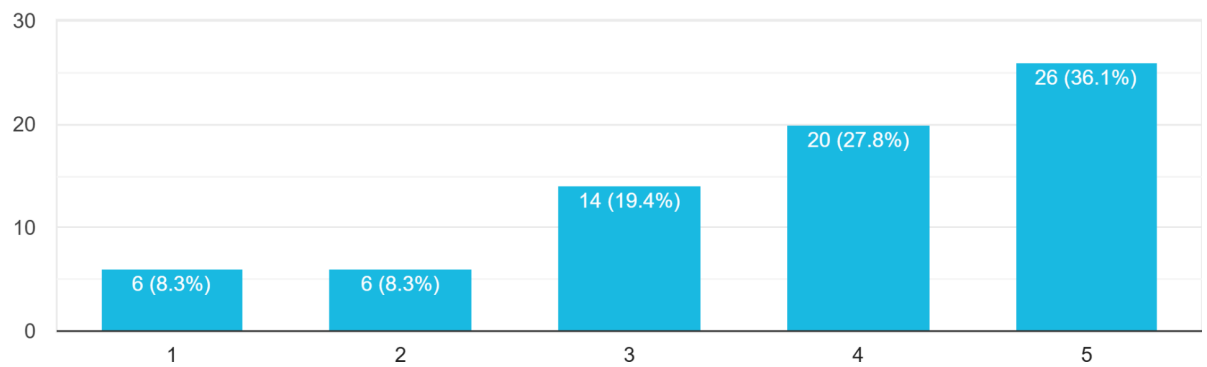
Are you interested in a built-in "Chat & Interview Scheduler" to reduce email clutter?

72 responses



How likely are you to switch to NexusPortal if it solved your verification and tracking issues?

72 responses



3.2.2 Sources of Information

- **Internal:** The shared experiences of Team_3 members facing scattered opportunities and missed deadlines.
- **External:** A Google Form questionnaire distributed to 68 participants to gather demographic and pain-point data.
- **Interviews:** Qualitative insights gathered from Faculty members regarding research lab management and employer needs for verified transcripts.
- **Documents:** Comparative analysis of existing job boards and academic portals to identify missing functional modules.

3.3 Gap Analysis

- **Current System:** Students struggle with fragmented listings across 4+ platforms and frequent "fake" or "ghost" postings. Faculty lack a centralized portal to find research assistants and must rely on trust rather than verification.
- **Required System (Nexus Portal):** A centralized digital ecosystem that provides a "University Verified" badge (desired by 82.4% of users), a dedicated research collaboration module, and an automated resume builder to reduce fraud and manual formatting.

3.4 Feature List Fixation

Based on survey analysis and stakeholder feedback, the following features are finalized for the initial product:

1. **Unified Dashboard:** A single feed for Jobs, Internships, and Research positions (82.4% user excitement).
2. **User Verification (ID & Transcript):** "University Verified" badges to build trust (82.4% trust increase).
3. **Resume Builder:** Auto-generates PDF CVs tailored to job descriptions (57.4% excitement).
4. **Research Collaboration Module:** A dedicated space for lab updates and RA recruitment (45.6% excitement).
5. **Application Tracker:** A Kanban-style board to monitor real-time application status (38.2% excitement).
6. **In-App Messaging & Interview Scheduler:** Integrated tools to reduce email clutter and coordinate meetings.
7. **Trust & Rating System:** A two-way review system for students, employers, and faculty (44.1% excitement).

8. **Skill-Gap Analyzer:** A tool to measure job readiness based on requirement matching.

4. Functional Requirements

4.1 Unified Opportunity Dashboard

- **Detailed Description:** The Unified Dashboard is the central landing page that aggregates three distinct types of opportunities: Corporate Jobs, Student Internships, and Academic Research Assistantships. It resolves the issue of fragmentation by presenting all relevant career options in a single, filterable feed. The system uses an algorithm to sort opportunities based on the user's major (e.g., CSE) and interests (e.g., Web Development, AI).
- **Inputs:**
 - **Search Queries:** Keywords (e.g., "React Developer", "Machine Learning RA").
 - **Filters:** Opportunity Type (Job/Internship/Research), Location (Remote/On-site), Stipend Range.
 - **User Profile Data:** Major and Skills (for auto-recommendations).
- **Outputs:**
 - **Opportunity Cards:** Summarized view containing Title, Organization/Faculty Name, Deadline, and "Verified" status.
 - **Detailed View:** Full description, requirements, and an "Apply Now" button when a card is clicked.
- **Interactions:**
 - Student selects a filter category $\rightarrow$ System updates the feed using AJAX.
 - Student clicks "Bookmark" $\rightarrow$ Opportunity is saved to the "Saved Items" list.

4.2 User Verification System

- **Detailed Description:** This feature implements a strict trust layer by verifying the academic identity of students and faculty. Users must upload official documents to earn a "University Verified" badge. This system addresses the 49.2% of users who have encountered fake job postings.
- **Inputs:**
 - **Document Upload:** Scanned image or PDF of a valid Student ID card or Academic Transcript.
 - **Admin Action:** Approval or Rejection decision with comments.
- **Outputs:**
 - **Verification Badge:** A visible green checkmark/badge next to the user's name on their profile and applications.
 - **Status Notification:** Email or system alert informing the user of the verification result.

- **Interactions:**
 - Student uploads ID $\rightarrow$ Status changes to "Pending Verification".
 - Admin reviews document and clicks "Approve" $\rightarrow$ System updates user role to "Verified Student".

4.3 Automated Resume Builder

- **Detailed Description:** The Resume Builder automatically pulls data from the student's verified profile (Education, Skills, Projects) to generate a standardized, industry-ready PDF resume. It eliminates the need for manual formatting in Word/Docs, which 43.1% of students currently struggle with.
- **Inputs:**
 - **Profile Data:** Fetched automatically from the database.
 - **Custom Fields:** Optional summary or specific objective text entered by the student.
 - **Template Selection:** Choice of "Academic" (Research-focused) or "Corporate" (Skill-focused) layout.
- **Outputs:**
 - **PDF File:** A downloadable, professionally formatted CV.
 - **Preview Mode:** A real-time visual preview of the document before downloading.
- **Interactions:**
 - Student clicks "Generate Resume" $\rightarrow$ System compiles data using PDFKit/Puppeteer.
 - Student edits a project description $\rightarrow$ Preview updates instantly.

4.4 Research Collaboration Module

- **Detailed Description:** A dedicated workspace for Faculty members to manage their research labs and recruit Research Assistants (RAs). Unlike standard job boards, this module includes tools specifically for academic workflows, such as code submission tracking and weekly progress updates.
- **Inputs:**
 - **Post Creation:** Research Topic, Required Skills (e.g., Python, MATLAB), Duration.
 - **RA Submission:** Code files, literature review documents, or progress logs.
- **Outputs:**
 - **Research Group Dashboard:** A private view accessible only to the Faculty lead and hired RAs.
 - **Contribution Logs:** A history of tasks completed by each student.
- **Interactions:**
 - Faculty creates a "Research Group" $\rightarrow$ System invites selected RAs.
 - RA submits a weekly update $\rightarrow$ Faculty receives a notification to review the work.

4.5 Application Tracking System (ATS)

- **Detailed Description:** A Kanban-style board that allows students to track the status of their applications in real-time. This feature directly addresses the frustration of 50.8% of students who rated tracking applications as a "nightmare" (Pain Level 4 or 5) .
- **Inputs:**
 - **Status Update:** Triggered by the Employer/Faculty (e.g., moving a candidate from "Applied" to "Interview").
- **Outputs:**
 - **Visual Board:** Columns for "Applied," "Under Review," "Interview," and "Offer/Rejected."
 - **Status Labels:** Color-coded tags (e.g., Yellow for Pending, Green for Accepted).
- **Interactions:**
 - Employer clicks "Shortlist" $\rightarrow$ Student's application card moves to the "Under Review" column on the student's dashboard.

4.6 In-App Messaging & Interview Scheduler

- **Detailed Description:** A secure communication channel that replaces scattered email threads. It allows recruiters and faculty to chat directly with applicants and schedule interviews using a built-in calendar integration.
- **Inputs:**
 - **Message Text:** Direct messages sent between verified users.
 - **Calendar Slot:** Date and time selected by the interviewer.
- **Outputs:**
 - **Chat History:** Persistent log of conversations.
 - **Meeting Invite:** A structured event card with a "Join Link" (for virtual interviews).
- **Interactions:**
 - Employer sends an interview request $\rightarrow$ Student receives a prompt to "Accept" or "Reschedule."
 - System sends a reminder notification 30 minutes before the scheduled interview.

5. External & Non-Functional Requirements

5.1 User Interfaces

- **Layouts:** The platform will feature a clean, professional dashboard designed for high scannability, prioritizing the Unified Feed of jobs, internships, and research opportunities.
- **Navigation:** A persistent sidebar or navigation drawer will allow users to switch rapidly between their Application Tracker, Resume Builder, and Research Collaboration modules.

- **Prototypes:** The interface will prominently display the "University Verified" badge on student and faculty profiles to foster immediate trust among stakeholders.
- **Responsive Design:** The UI will be fully responsive to ensure seamless access across desktops and mobile devices, catering to the "on the go" needs of students.

5.2 Software Interfaces

- **Integration with other software:** The Resume Builder will interface with PDF generation libraries to export standardized, industry-ready CVs.
- **Communication Services:** Integration with SMTP services for automated notifications and a secure socket-based interface for the In-App Messaging system.
- **Verification APIs:** Future iterations plan to interface with University Database APIs for automated transcript and enrollment verification.

5.3 Performance Requirements

- **Speed:** The system must handle real-time status updates in the Application Tracker with minimal latency to ensure students do not miss critical deadlines.
- **Scalability:** The architecture will support an initial load of 4.2 Million+ tertiary students and 15,000+ faculty members, primarily within the Dhaka metropolitan area, with capacity for nationwide expansion.
- **Reliability:** Optimized database queries and high-availability server configurations will be used to handle high-frequency searches across complex multi-module listings.

5.4 Security Requirements

- **Authentication:** Multi-factor authentication (MFA) and strict role-based access control (RBAC) for Students, Faculty, Employers, and the Admin Team.
- **Data Protection:** Implementation of industry-standard encryption for sensitive academic documents, ID scans, and transcripts to prevent unauthorized access.
- **Access Control:** An Admin Verification layer will manually and automatically check documents to prevent fake or "ghost" postings, which currently affect 18.5% of surveyed users.

5.5 Maintainability

- **Ease of modification:** The system will be built using a modular approach to allow for the easy addition of future features such as AI-assisted career tools and Skill-Gap Analysis.
- **Ease of extension:** Comprehensive code documentation and API schemas will be maintained to facilitate the transition to a Mobile App in the second release phase.
- **Monitoring:** Integrated error logging and performance monitoring to proactively identify and fix bugs in the complex multi-module ecosystem.
-

6. Feasibility & Project Management Analysis

6.1 Feasibility Analysis

6.1.1 Technical Feasibility

The Nexus Portal system is technically feasible using current and well-established web technologies. The proposed architecture follows a modular, role-based design that separates student, faculty, employer, and admin functionalities, making development and maintenance manageable. Core features such as dashboards, verification workflows, application tracking, resume generation, and messaging can be implemented using standard web frameworks, relational databases, and secure APIs.

The system does not require experimental or unavailable technologies; therefore, the technical risks are minimal and manageable within an academic project scope.

6.1.2 Economic / Financial Feasibility

From a financial perspective, Nexus Portal is economically feasible. The system can be developed using open-source tools, frameworks, and libraries, which significantly reduces development costs. As an academic project, no immediate infrastructure or licensing investment is required.

In a real-world deployment scenario, the platform could adopt sustainable revenue models such as institutional subscriptions, premium employer postings, or verified badge services, making it financially viable in the long term.

6.1.3 Operational Feasibility

Operational feasibility is high, as the system is designed to align with existing academic and recruitment workflows. Students are already accustomed to online job portals, faculty regularly recruit research assistants, and employers actively use digital hiring platforms.

The introduction of a centralized, verified system simplifies these processes rather than disrupting them. The admin-controlled verification and moderation mechanisms ensure smooth day-to-day operation and maintain platform credibility.

6.1.4 Human Resources Feasibility

The project is human-resource feasible given the team's academic background and skill set. The development tasks are divided among team members based on roles such as frontend design, system analysis, and feature modeling.

No specialized or external expertise is required beyond standard software engineering and system analysis knowledge covered in the course curriculum, making the project achievable within the team's capabilities.

6.1.5 Market / Legal / Political Feasibility

From a market perspective, Nexus Portal addresses a clear and validated demand, as demonstrated by survey responses highlighting opportunity fragmentation and trust issues.

Legally, the system adheres to basic data protection principles by incorporating role-based access control and secure document handling. There are no political or regulatory barriers affecting the implementation of a university-focused recruitment platform, making this aspect feasible with proper administrative oversight.

6.1.6 Schedule Feasibility

The project is schedule feasible within the academic timeline of the SAD Lab course. The system is developed in clearly defined phases, including requirement analysis, system design, prototyping, and documentation.

By prioritizing core features for the initial version and following an incremental development approach, the project can be completed within the allotted semester timeframe.

6.2 SWOT Analysis

Strengths

- Unified platform combining jobs, internships, and research opportunities.
- “University Verified” badge increases trust and reduces fraud.
- Dedicated research collaboration module not available in traditional job portals.
- Role-based system supporting students, faculty, employers, and admins.
- Integrated resume builder and application tracking improve user experience.
- Strong admin control for moderation and verification.

Weaknesses

- Heavy reliance on manual admin verification in the initial phase.
- Limited real-world integration with official university databases at launch.
- Adoption depends on institutional participation and user onboarding.
- Feature-rich system may increase initial learning curve for users.

Opportunities

- Expansion to nationwide or international universities.
- Integration with official university APIs for automated verification.
- Addition of AI-assisted career guidance and skill-gap analysis.
- Potential partnerships with academic institutions and industry recruiters.

- Mobile application development in future phases.

Threats

- Competition from established platforms like LinkedIn or Bdjobs.
- Data privacy and security risks if not properly managed.
- Resistance from institutions hesitant to adopt new systems.
- Dependence on admin efficiency to maintain platform credibility.
-

7. System Design

7.1 Data Flow Diagram

The Level-1 DFD breaks the NexusPortal system into seven core processes and five primary data stores to show how information moves from user registration and verification to final recruitment and research collaboration.

Processes:

- **P1: User Authentication & Profile Management:** Handles secure login, role selection (Student/Faculty/Employer), and profile data updates.
- **P2: Verification System:** Processes ID and Transcript uploads, routes them to Admins for review, and assigns "University Verified" badges.
- **P3: Opportunity Management:** Aggregates Jobs, Internships, and Research Assistantships into a single Unified Dashboard.
- **P4: Application Tracking System (ATS):** Manages the lifecycle of applications from submission to "Interview" and "Hired" statuses.
- **P5: Resume Builder:** Extracts verified profile data to auto-generate professional PDF resumes.
- **P6: Research Collaboration Module:** Facilitates lab management, code submissions, and Research Assistant (RA) contribution tracking.
- **P7: In-App Messaging:** Enables secure, direct communication and interview scheduling between applicants and recruiters.

Data Stores: **D1:** User Database (Profiles/Auth), **D2:** Opportunity Database (Jobs/Research), **D3:** Application Database (Status/Tracking), **D4:** Research Group Database (Labs/RAs), **D5:** Document Storage (IDs/Resumes).

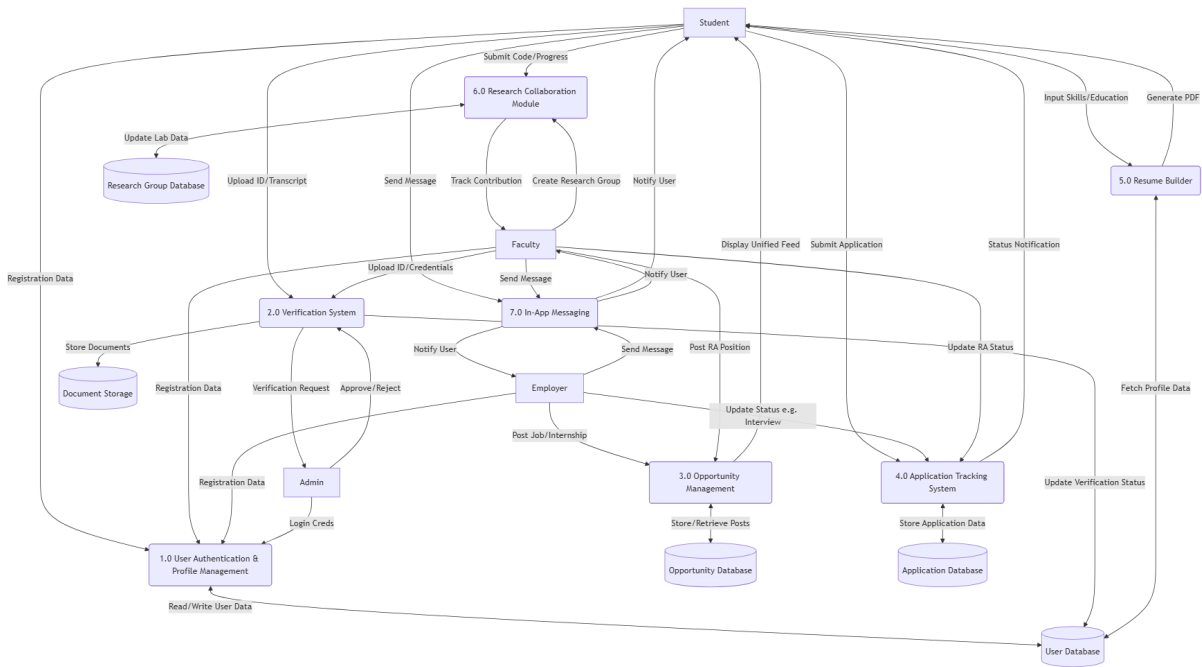


Figure: Level-1 Data Flow Diagram

7.2 Use Case Diagram

The Use Case Diagram defines the functional requirements by showing the relationships between actors and system actions.

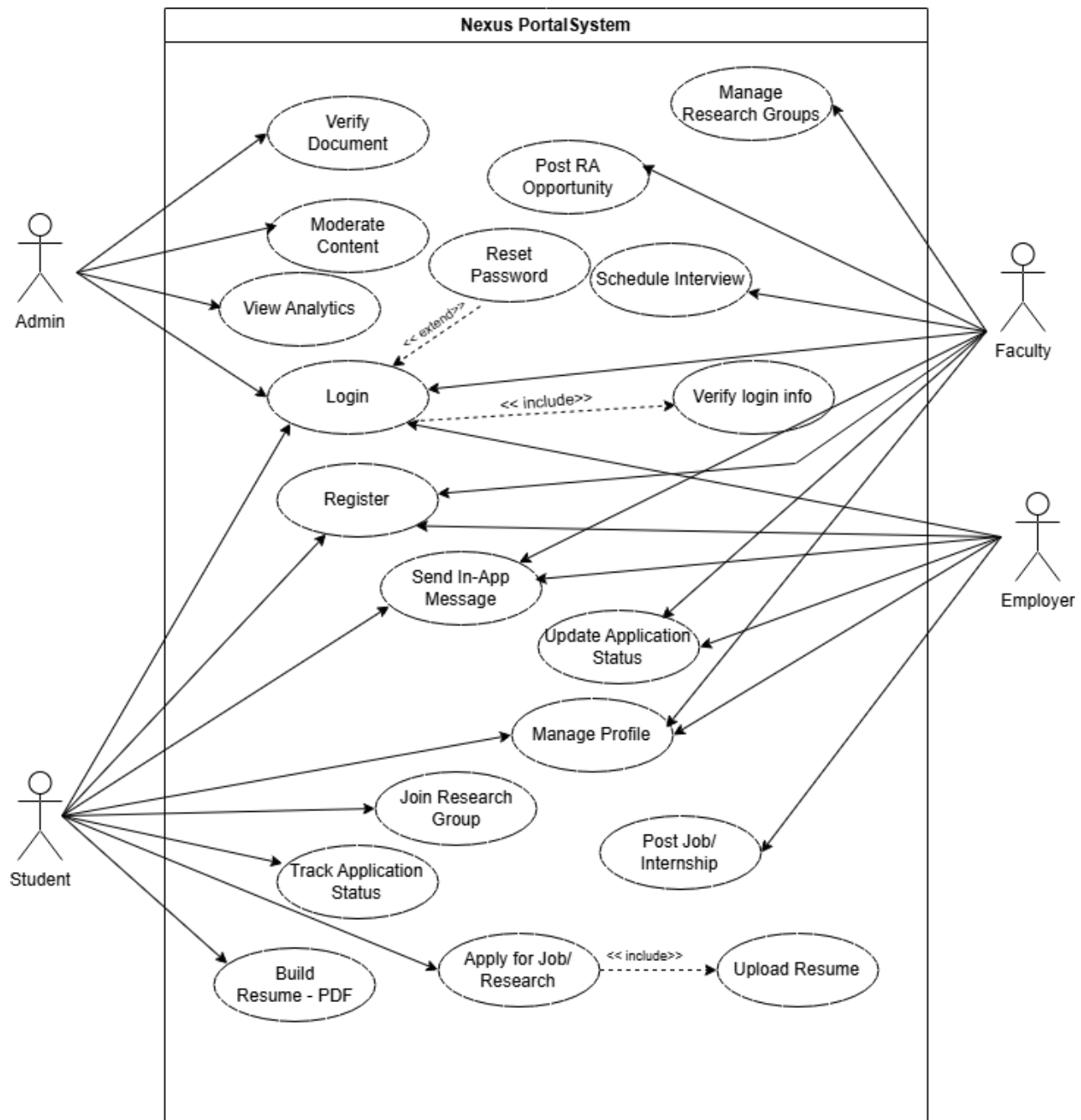


Figure: Use Case Diagram

7.3 Class Diagram

The Class Diagram provides a static, structural view of the NexusPortal system. It outlines the key classes such as User (specialized into Student, Faculty, Employer, and Admin), Opportunity, Application, Resume, ResearchGroup, and VerificationDocument, along with their attributes (data) and methods (operations). It also illustrates the relationships and cardinalities between these classes, for example, how a Student can submit multiple Applications, generate a standardized Resume, and upload Documents to earn a "University Verified" badge. Additionally, the diagram shows how Faculty are linked to ResearchGroups to manage laboratory collaborations, and how Employers interact with the system to post Opportunities and schedule interviews. This diagram acts as a blueprint for both the database schema and the object-oriented system structure, ensuring smooth integration between the unified opportunity feed, application tracking, and research collaboration modules.

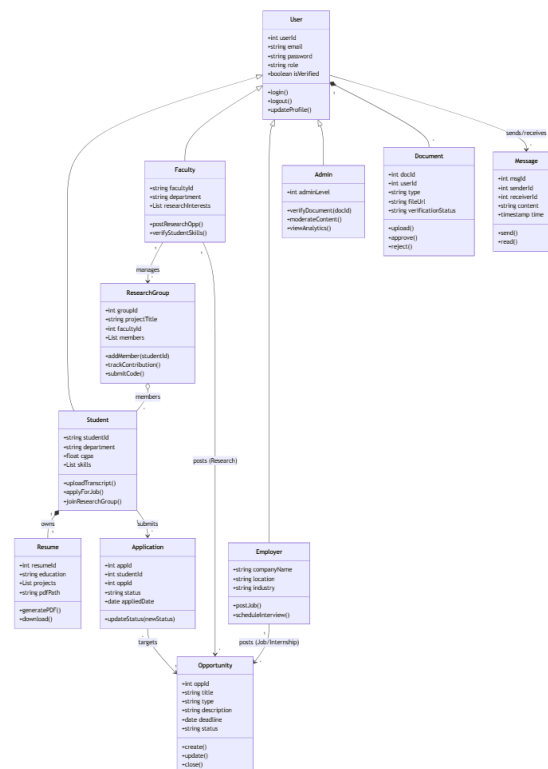


Figure: Class Diagram

7.4 Sequence Diagram

This section details the dynamic interactions between users (Actors) and the system components (Frontend, Backend, Database) for two critical workflows: applying for an opportunity and verifying a user account.

7.4.1 Sequence Diagram: Student Job Application

Process Description:

1. **Initiation:** The Student interacts with the NexusPortal UI by clicking "Apply Now" on a job listing.
2. **Verification Check:** The Frontend sends a request to the System Server, which immediately queries the Database to ensure the student's status is "VERIFIED."
3. **Record Creation:** Once verified, the Database creates a new Application Record (e.g., ID: 501) and confirms the save to the Server.
4. **Notification:** The System Server sends a real-time notification to the Employer regarding the new applicant. The Employer (or their system) acknowledges receipt.
5. **Completion:** The Server confirms the successful transaction to the UI, which displays a "Success" message to the Student, completing the loop.

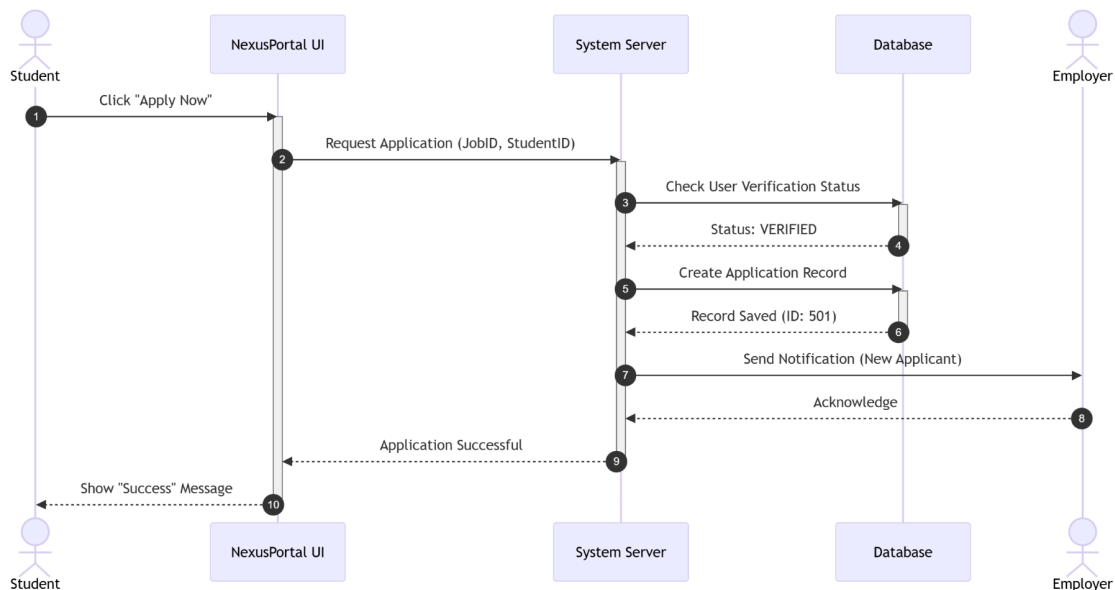


Figure: Sequence Diagram - 1

7.4.2 Sequence Diagram: Admin Verification Workflow

Process Description:

1. **Retrieval:** The Admin accesses the Dashboard to view a list of "Pending Verifications." The Backend fetches this list from the database.
2. **Review:** The Admin selects a specific student (e.g., Student_A) and requests to view their uploaded transcript. The Backend retrieves the secure PDF document for the Admin to inspect manually on the screen.
3. **Decision:** Upon confirming the document's validity, the Admin clicks "Approve Verification."
4. **Update:** The Backend sends an UPDATE command to the Database, changing the user's status to `isVerified = TRUE`.
5. **Notification:** The system automatically notifies the Student that their account is now verified.
6. **Feedback:** The Admin Dashboard receives a confirmation toast message indicating the update was successful.

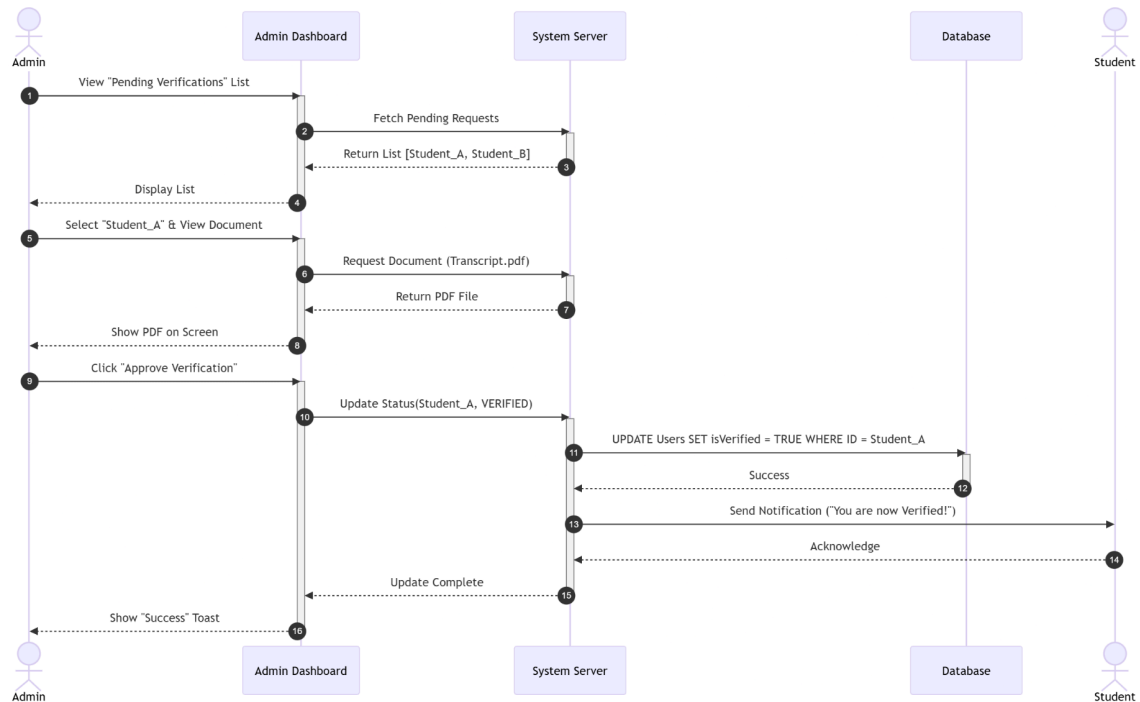
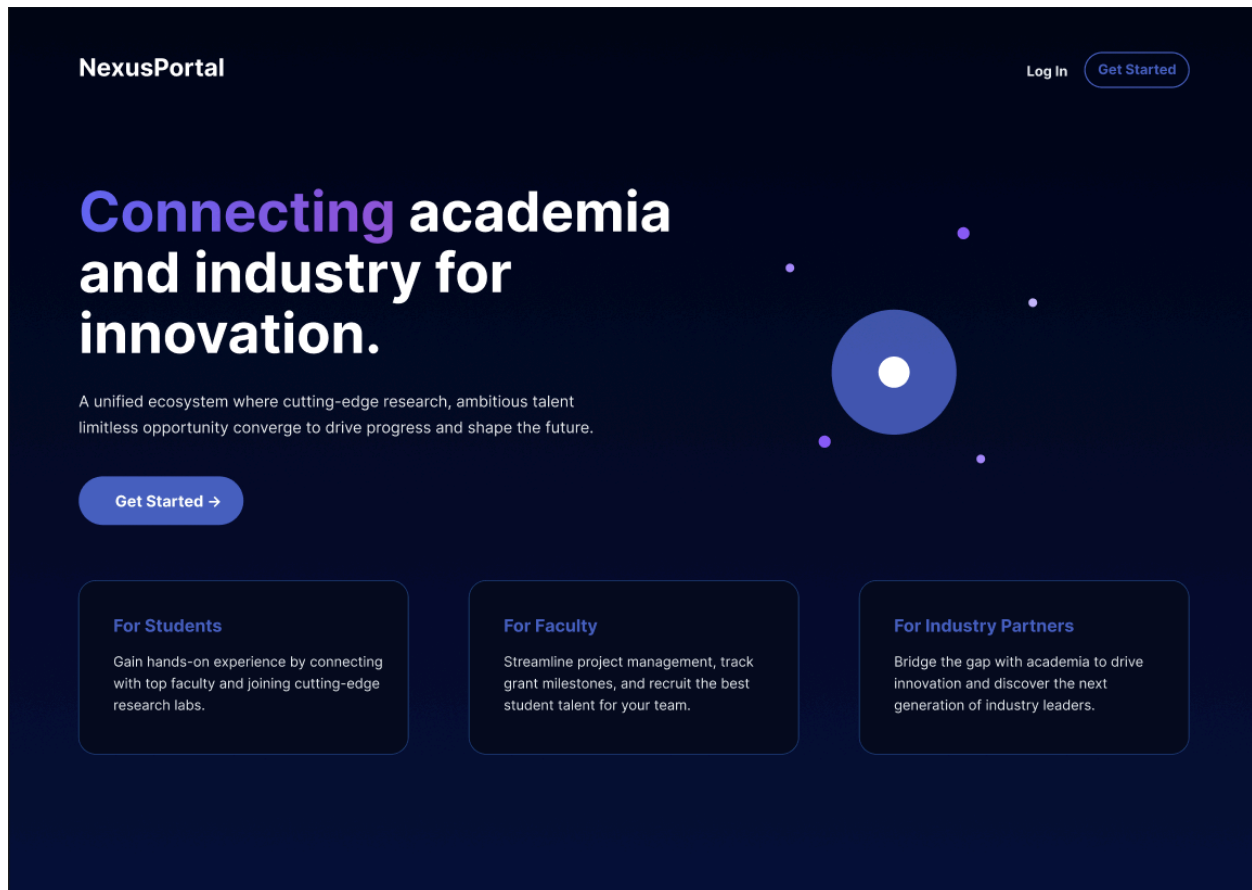


Figure: Sequence Diagram - 2

8. Prototype Design

8.1 Screenshots of prototypes



Landing Page

Welcome back

Sign in to continue to your account

Email address

Password

[Forgot password?](#)

☐ Remember me for 30 days

[Sign in](#)

or continue with



Don't have an account? [Sign up](#)



Your Academic & Career Hub

Connect with opportunities, collaborate on research, and accelerate your career journey.

Login page

Create your account

Join NexusPortal and start your journey

I am a...


Employee


Faculty


Employer

First name

John

Last name

Doe

Email address

john@university.edu

Password

Create a strong password

Must be at least 8 characters with a number and special character

Create account

By signing up, you agree to our [Terms of Service](#) and [Privacy Policy](#)

or continue with

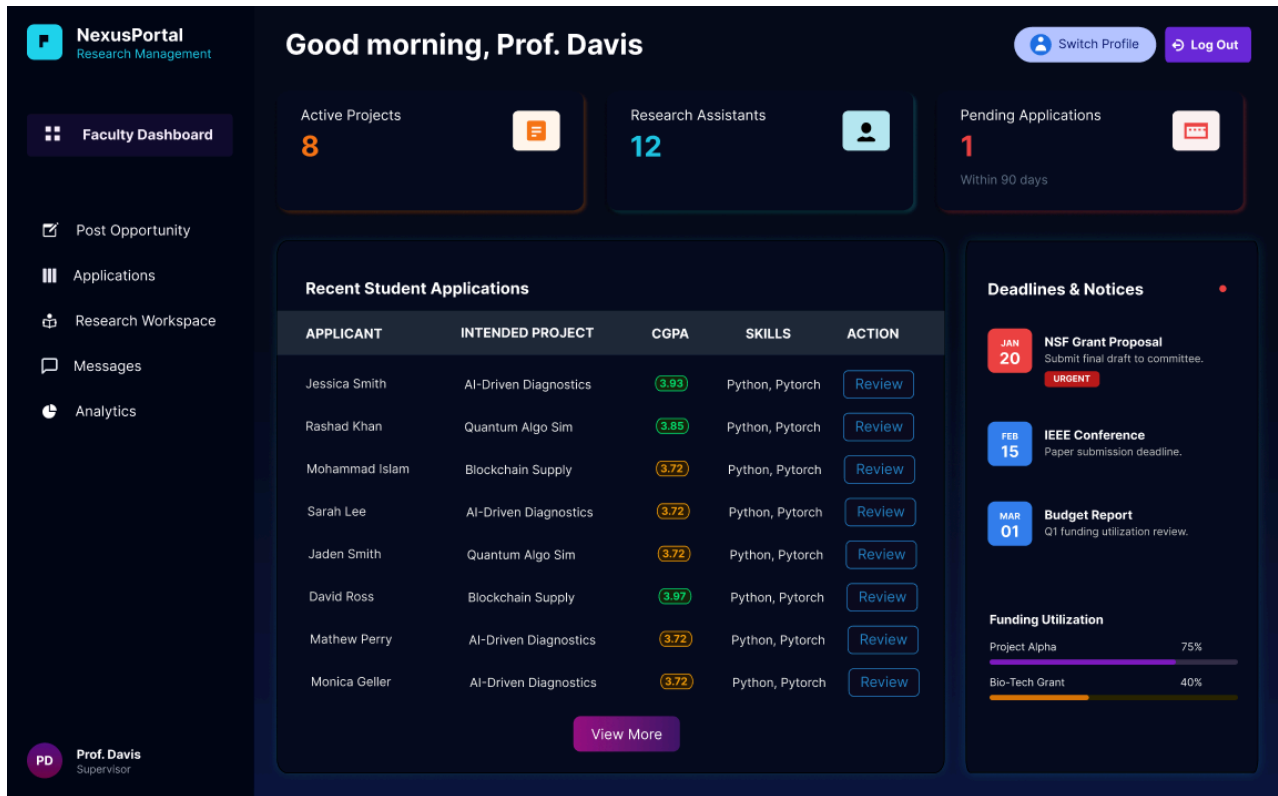






Already have an account? [Sign in](#)

Get Started page



Faculty Dashboard page

 **NexusPortal**
Research Management

Faculty Dashboard

Post Opportunity

Applications

Research Workspace

Messages

Analytics

PD

Prof. Davis
Supervisor

Switch Profile

Log Out

Post a Research Opportunity

Create a new research position to attract talented students.

Basic Information

Provide the main details about the research opportunity.

Position Title *

e.g., Research Assistant - Machine Learning Lab

Department *

Select department

Opportunity Type *

Select type

Description *

Describe the research opportunity, objectives, and what the student will be working on...

Requirements & Skills

Specify what you're looking for in candidates.

Academic Level *

Select level

Minimum GPA

e.g., 3.5

Required Skills

Faculty Post Opportunity page

40

NexusPortal

Research Management

Switch Profile

Log Out

Faculty Dashboard

Post Opportunity

Applications

Research Workspace

Messages

Analytics

PD Prof. Davis
Supervisor

Applications

Review and manage student applications for your research positions.

Search applications...

All Projects

+ All (5)

Pending (2)

Interviewing (1)

Accepted (1)

Rejected (1)

Emily Chen

emily.chen@university.edu

Research Assistant - AI Drug Discovery

Computer Science • Junior GPA: 3.9 ★ 4.5

Machine Learning Bioinformatics Python

Resume Contact View

✓ ✕

Pending

Marcus Johnson

m.johnson@university.edu

Research Assistant - Quantum Computing

Physics • Senior GPA: 3.7 ★ 4.2

Quantum Computing Algorithm Research

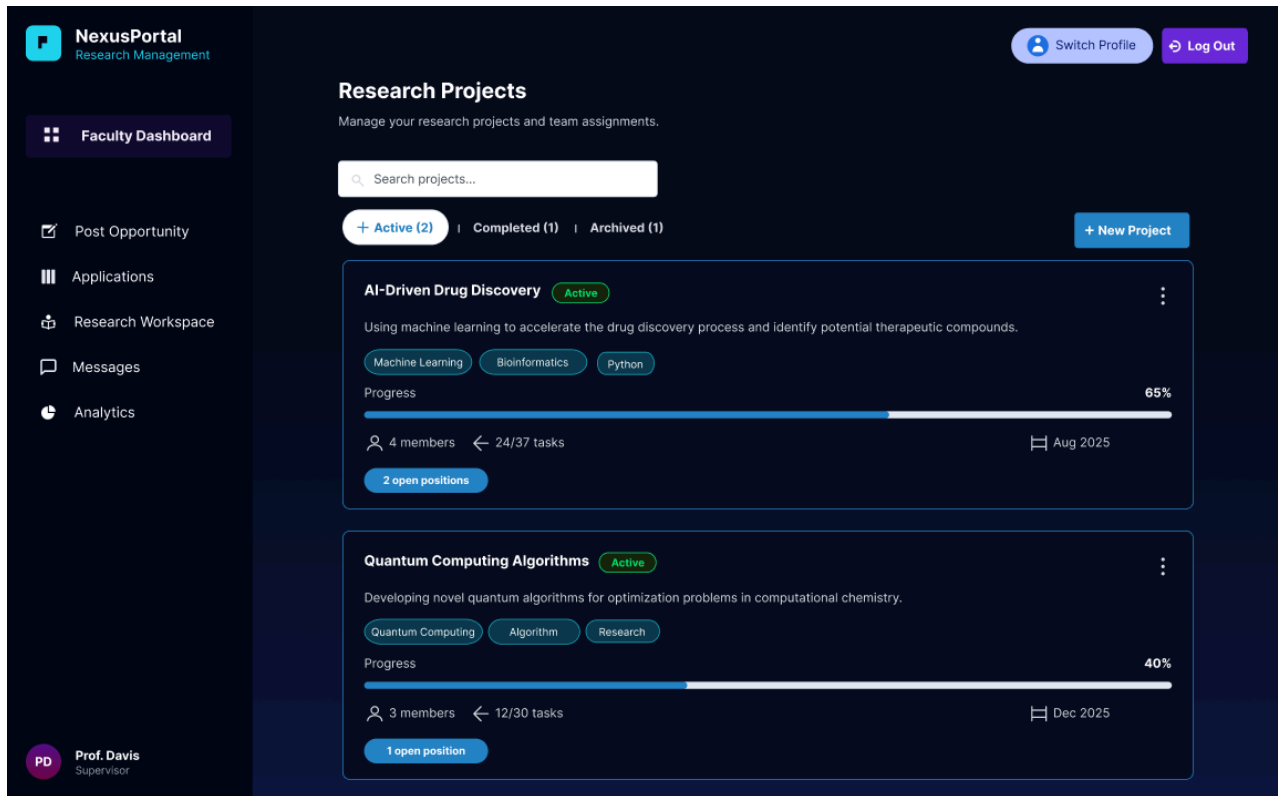
Resume Contact View

✓ ✕

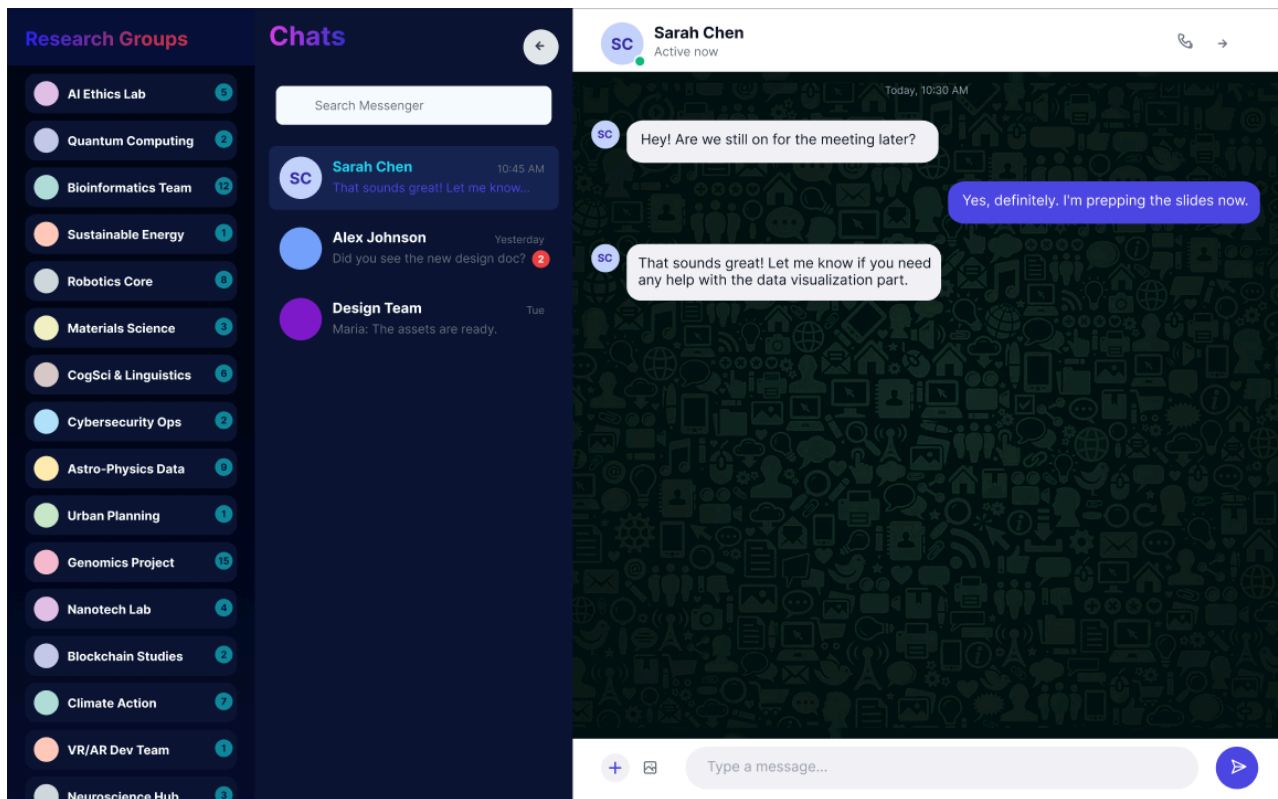
Pending

Faculty View Applications page

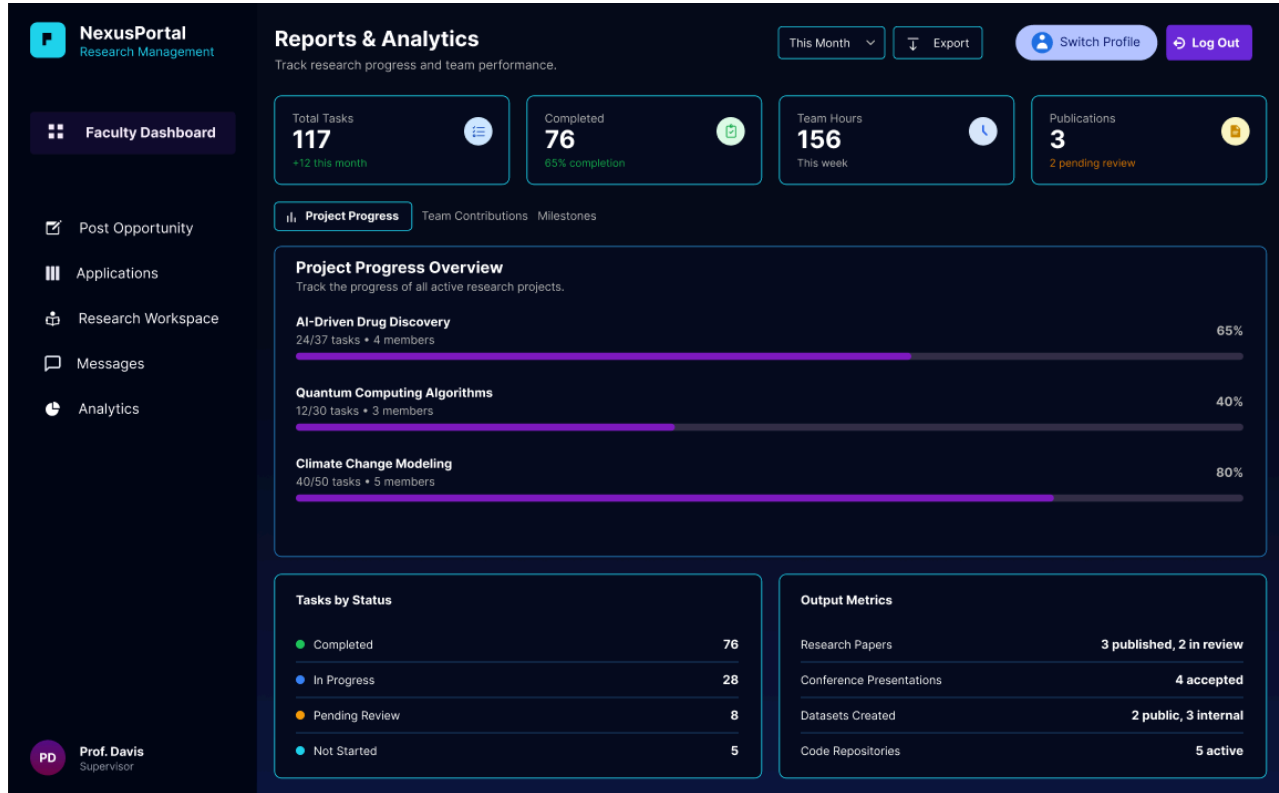
41



Faculty Research projects page



Messages page



Faculty Reports and analytics page

NexusPortal

Research Management

Faculty Dashboard

CURRENT PROJECT

AI in Climate Modeling

NSF Grant • Active

Milestones

Tasks4

Documents

Publications2

Messages

PD

Prof. Davis

Supervisor

Good morning, Prof. Davis

You have 5 items requiring attention across 3 projects.

Pending Reviews

3

Your action required

Upcoming Deadlines

8

Within 7 days

Grants Expiring

1

Within 90 days

My Project Portfolio

View all

AI in Climate Modeling

NSF Grant #2024-1234

→ On Track

Progress

60%

Data Cleanup Phase

3 days

Genomic Sequencing Study

NIH R01 #GM-5678

At Risk

Progress

25%

Lab Access Approval

1 days

Urban Mobility Patterns

DOT Grant #TR-9012

→ On Track

Progress

45%

Survey Distribution

12 days

Unified Task Inbox

Due Soon

Review Methodology ...

AI Climate Modeling

Waiting on you

Today

Submit Ethics Renewal

Genomic Sequencing Study

Tomorrow

Approve budget for lab equipment

AI Climate Modeling

Fri

Review student thesis chapter 3

Genomic Sequencing Study

Next Week

Schedule team sync meeting

Urban Mobility Patterns

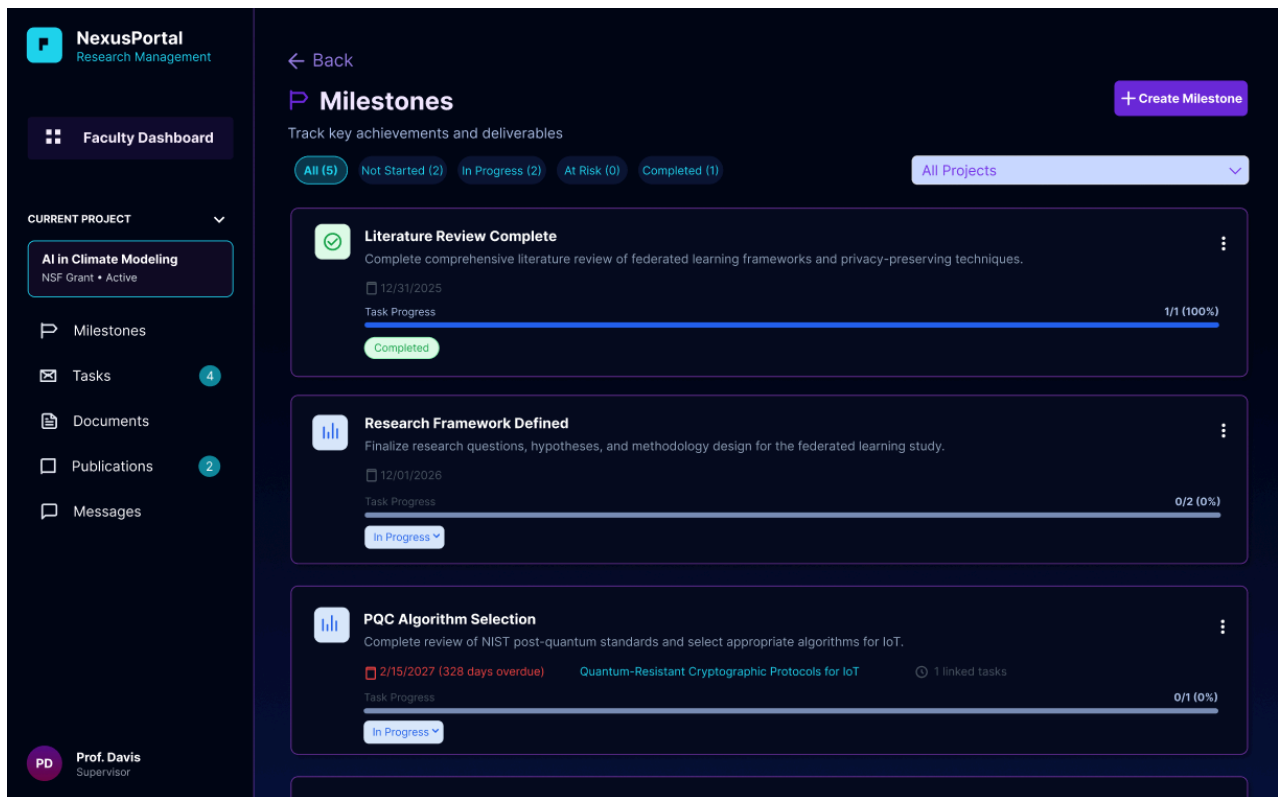
Next Week

View all tasks

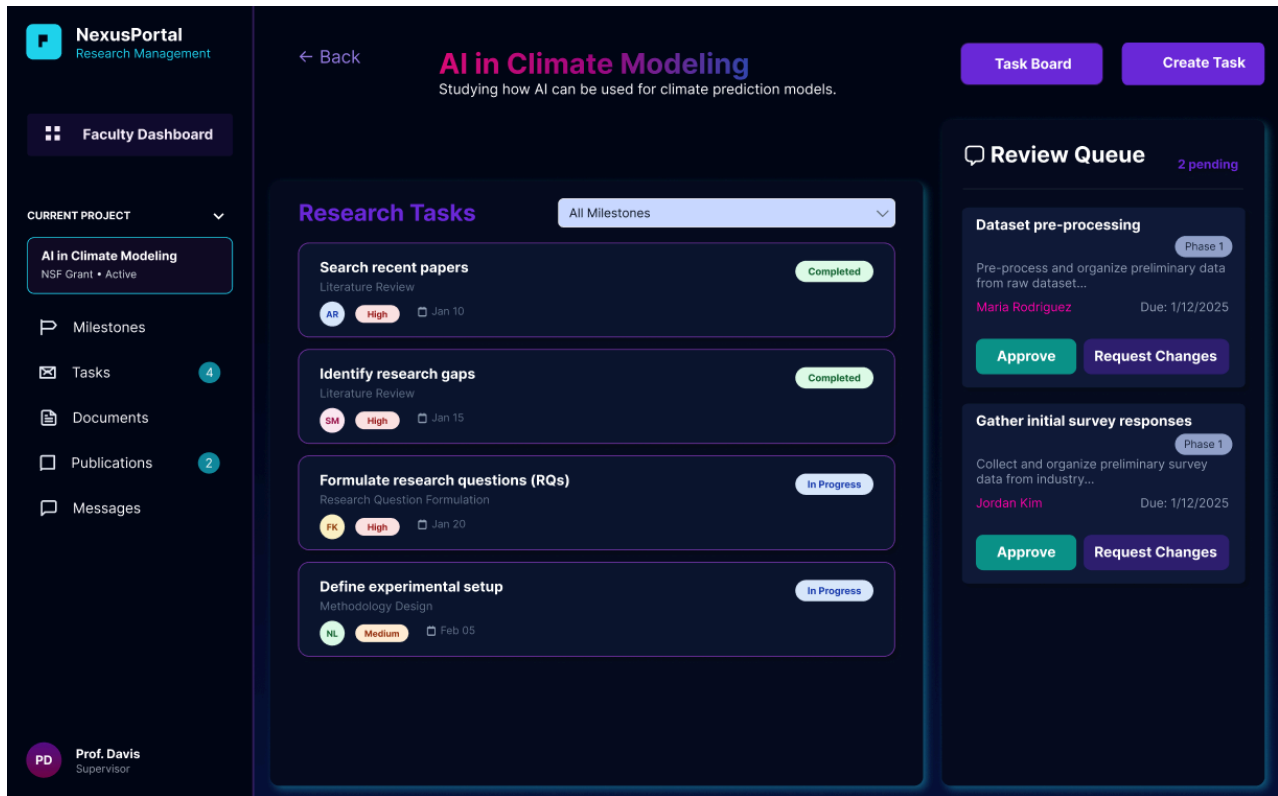
Faculty Research Space



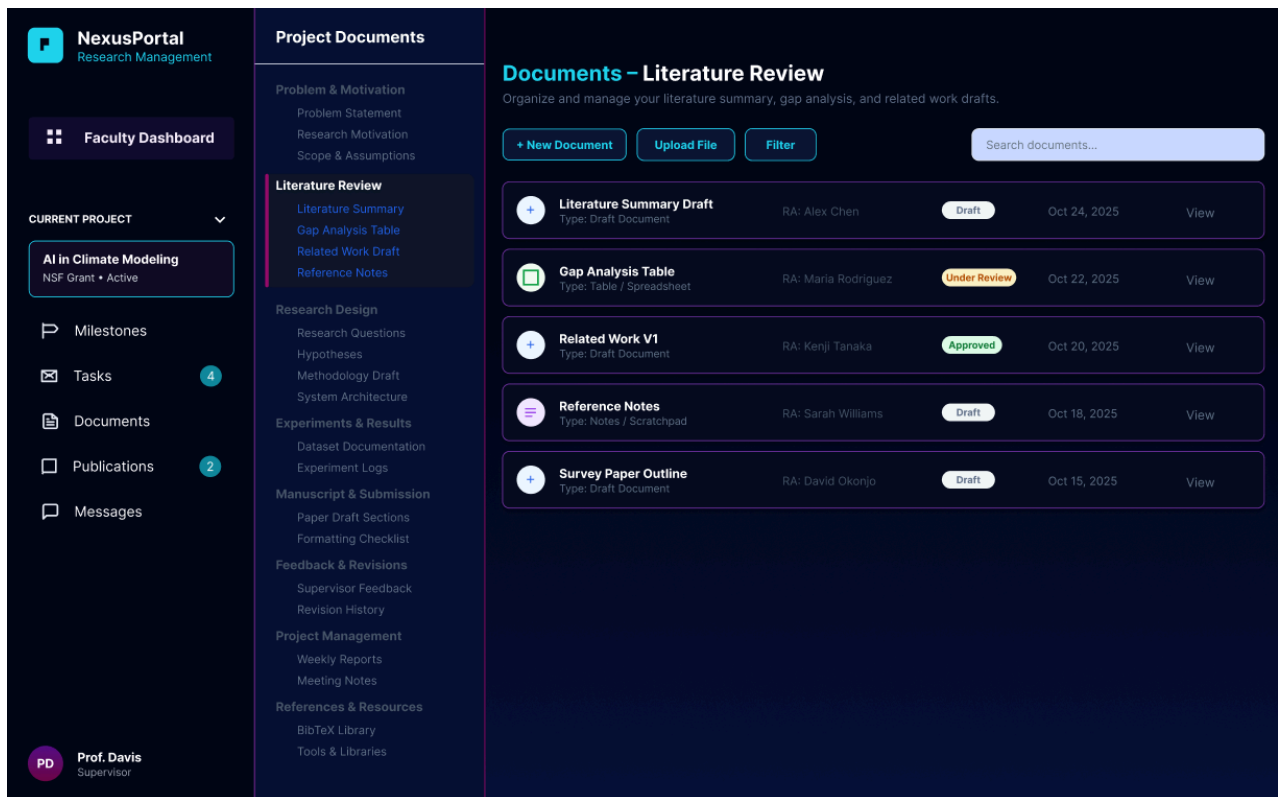
Faculty Project View page



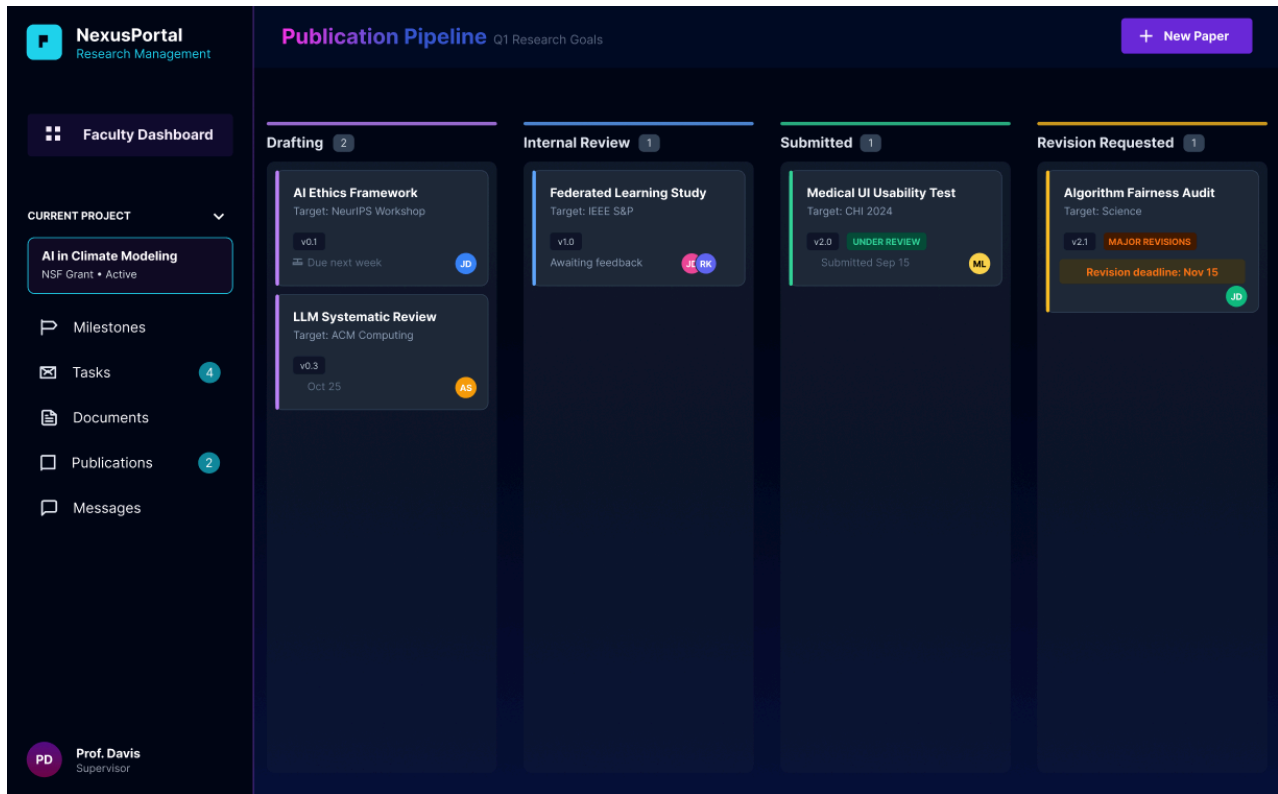
Faculty Milestones View page



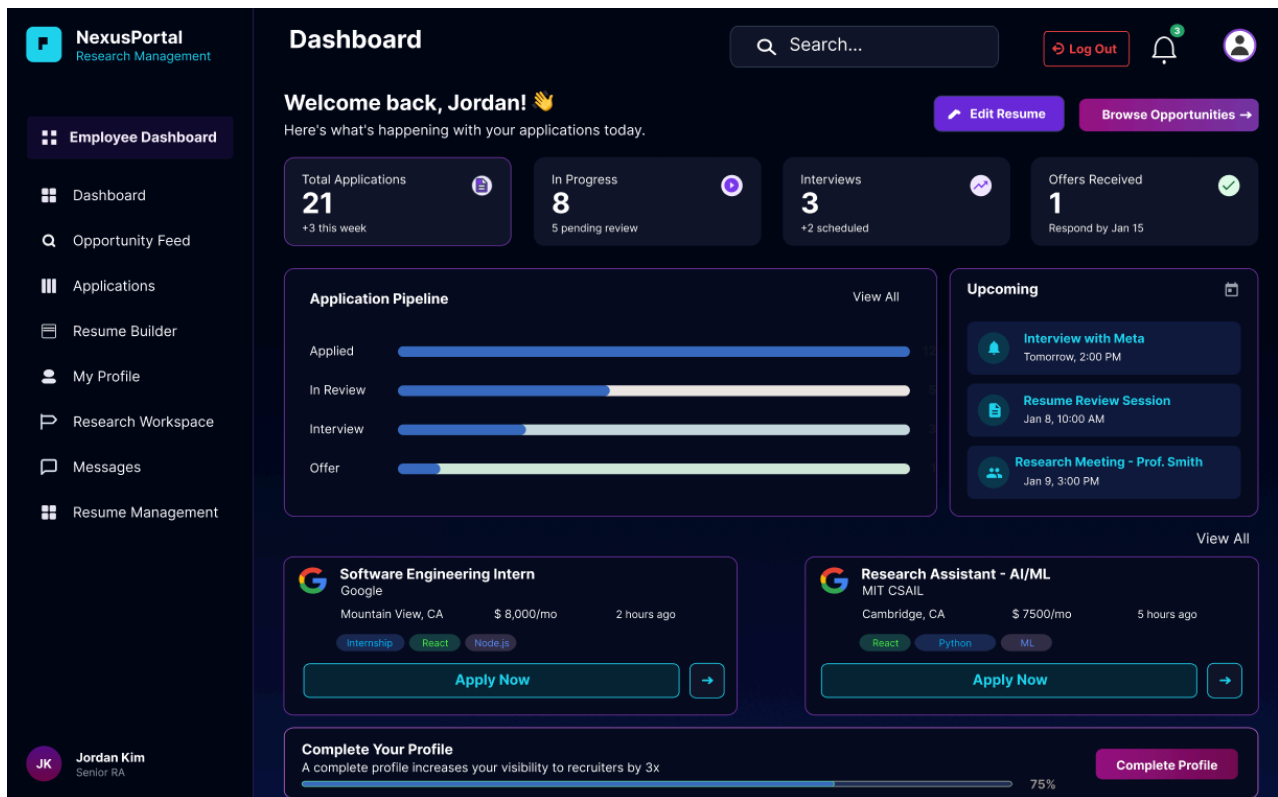
Faculty Tasks View page



Faculty Research Documents View page



Faculty Publications page



Employee/Student Dashboard page

NexusPortal
Research Management

Resume Builder

Employee Dashboard

Dashboard

Opportunity Feed

Applications

Resume Builder

My Profile

Research Workspace

Messages

Resume Management

JK

Jordan Kim

Senior RA

Personal Information

First Name

Alex

Last Name

Johnson

alex.johnson@university.edu

Phone

(555) 123-4567

Location

San Francisco, CA

Work Experience

Software Engineering Intern

TechCorp Inc.

Jun 2024

Aug 2024

+ Add Experience

Education

Stanford University

B.S. Computer Science

GPA

3.8/4.0

+ Add Education

JavaScript

TypeScript

React

+ Add skill...

+ Add Project

+ Add Section

Modern Template

+ AI Enhance

Save

Export PDF

Alex Johnson

San Francisco, CA

alex.johnson@university.edu

(555) 123-4567

linkedin.com/in/alexjohnson

Work Experience

Software Engineering Intern

Jun 2024 - Aug 2024

TechCorp Inc.

- Developed and maintained React components for the customer dashboard
- Implemented REST API integrations reducing load time by 40%
- Collaborated with design team to improve UX/UI

Education

Stanford University

2021 - 2025

B.S. Computer Science • GPA: 3.8/4.0

Skills

JavaScript, TypeScript, React, Node.js, Python, SQL, Git, AWS

Projects

E-commerce Platform

React, Node.js, PostgreSQL

Built a full-stack e-commerce platform with payment processing and inventory management

Employee/Student Resume builder page

NexusPortal
Research Management

Opportunity Feed

Location

Employee Dashboard

Dashboard

Opportunity Feed

Applications

Resume Builder

My Profile

Research Workspace

Messages

Resume Management

JK

Jordan Kim

Senior RA

Full-time

Part-time

Internship

Research

Contract

Showing 6 opportunities

Software Engineering Intern

Google

Mountain View, CA

\$ 8,000/mo

2 hours ago

Internship

React

TypeScript

Node.js

Apply Now

Research Assistant - AI/ML

MIT CSAIL

Cambridge, MA

\$ 7,500/mo

5 hours ago

Research

Python

Machine Learning

Apply Now

Junior Frontend Developer

Stripe

Mountain View, CA

\$ 8,000/mo

2 hours ago

Internship

React

Apply Now

Data Science Intern

Meta

Menlo Park, CA

\$ 7,500/mo

2 days ago

Internship

Python

SQL

Apply Now

Graduate Research Assistant

Stanford University

Stanford, CA

\$ 7,500/mo

3 days ago

Research

Apply Now

Backend Engineer - Part Time

Notion

Remote

\$ 60/hr

4 days ago

Part-time

Apply Now

Load More Opportunities

Employee/Student Opportunity feed page

48



NexusPortal

Research Management

Application Tracker

Search




Track all your applications in one place. Drag cards to update their status.

Search applications...

Filter

Add Application

Employee Dashboard

Dashboard

Opportunity Feed

Applications

Resume Builder

My Profile

Research Workspace

Messages

Resume Management

Saved

2

Product Designer

Figma • Remote

Full-time

Saved 3 days ago

UX Research Intern

Airbnb • SF, CA

Internship

Saved 1 week ago

Applied

3

Software Eng Intern

Google • Mountain View

Internship

Applied Jan 2

Frontend Developer

Stripe • SF, CA

Full-time

Applied Dec 28

Data Science Intern

Meta • Menlo Park

Internship

Applied Dec 25

Screening

1

ML Engineer

OpenAI • SF, CA

Full-time

Screening since Jan 1

Interview

2

Backend Engineer

Notion • Remote

Full-time

Interview: Tomorrow

Research Assistant

Stanford NLP Lab

Research

Interview: Jan 8

Offer

1

Software Development

Amazon • Seattle

Full-time

\$150k

Respond by Jan 15

Saved

2 applications

Applied

3 applications

Screening

1 application

Interview

2 application

Software Development

1 application

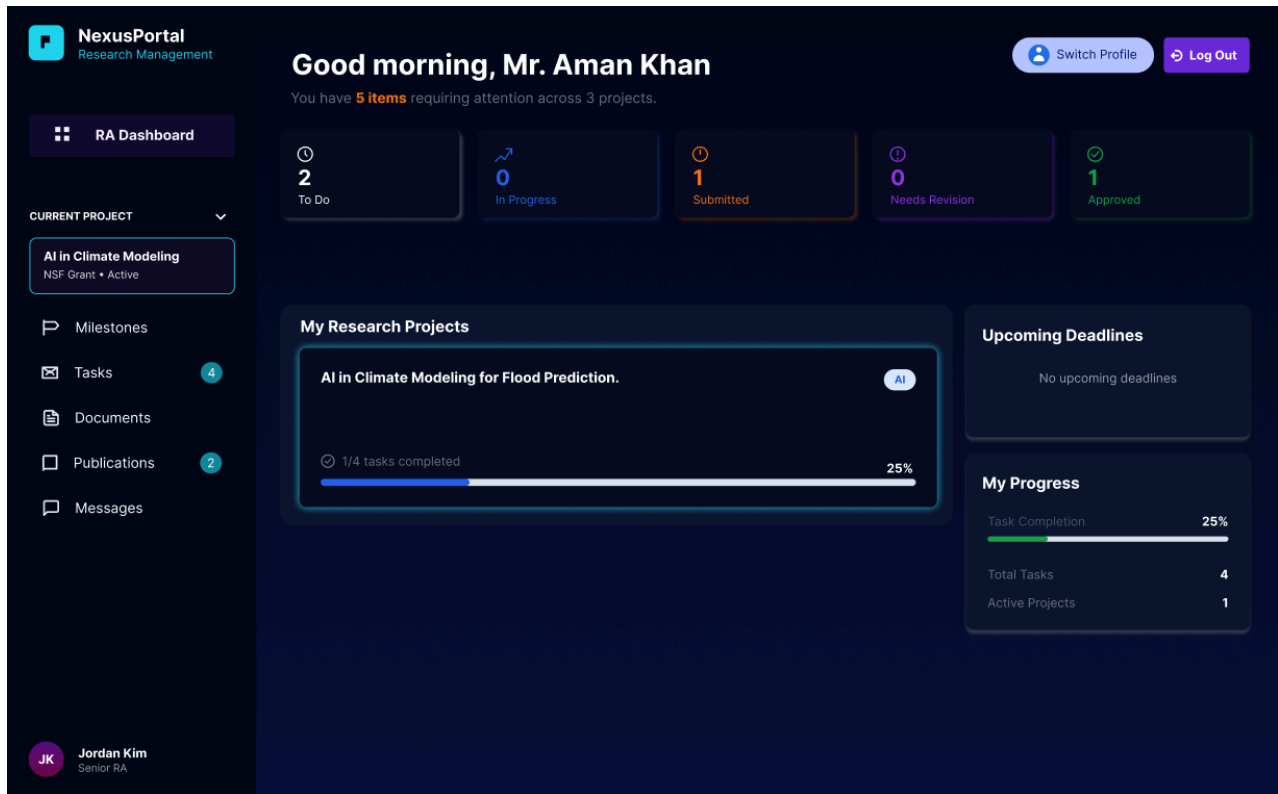
JK

Jordan Kim

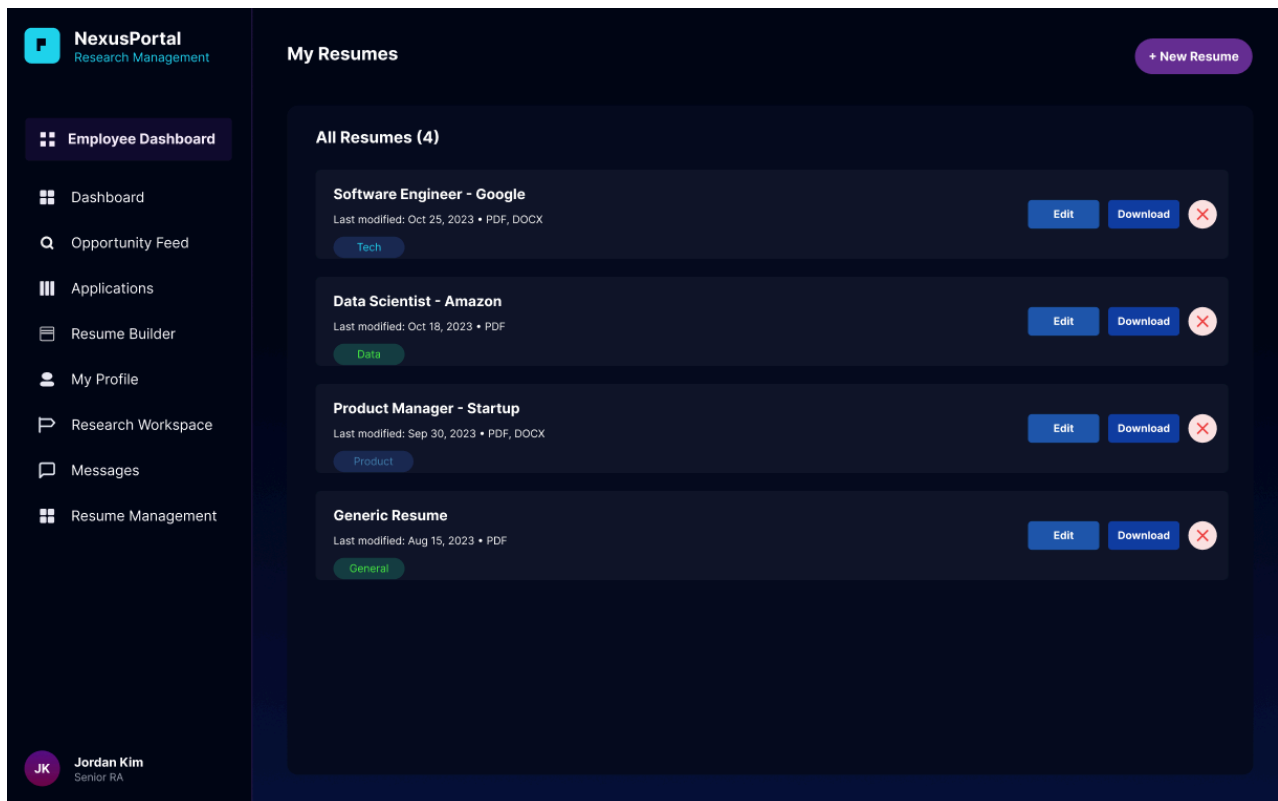
Senior RA

Employee/Student Application tracker page

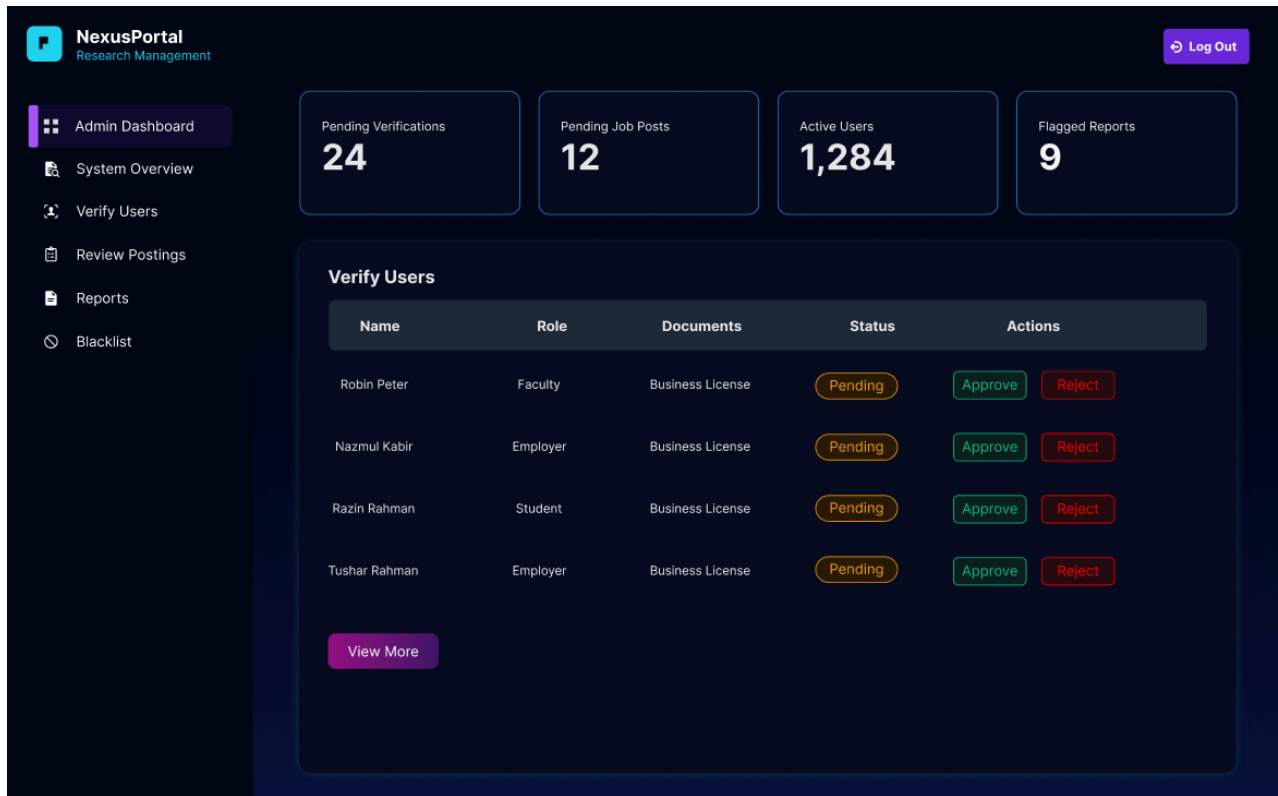
49



Employee/Student Research Assistant Dashboard page



Employee/Student Resume Management



Admin Dashboard page



Admin System Overview page


NexusPortal
 Research Management

- Admin Dashboard
- System Overview
- Verify Users**
- Review Postings
- Reports
- Blacklist

Role: All
 Status: Pending

Name	Role	Documents	Status	Actions
Robin peter	Faculty	Business License	Pending	Approve Reject
Nazmul Kabir	Employer	Business License	Pending	Approve Reject
Razin Rahman	Student	Business License	Pending	Approve Reject
Tushar Rahman	Employer	Business License	Pending	Approve Reject
Priya	Student	Business License	Pending	Approve Reject
Robin peter	Faculty	Business License	Pending	Approve Reject
Nazmul Kabir	Employer	Business License	Pending	Approve Reject
Razin Rahman	Student	Business License	Pending	Approve Reject
Tushar Rahman	Employer	Business License	Pending	Approve Reject
Priya	Student	Business License	Pending	Approve Reject

Admin Verify users page


NexusPortal
 Research Management

- Admin Dashboard
- System Overview
- Verify Users
- Review Postings**
- Reports
- Blacklist

Category: All
 Status: Pending

Title	Posted By	Category	Status	Actions
Software Engineer Intern	Google	Internship	Pending	Approve Reject Flag
AI Research Assistant	MIT CSAIL	Research	Pending	Approve Reject Flag
FrontEnd Developer	Shohoz	Job Post	Pending	Approve Reject Flag
DSA Research Assistant	Harvard	Research	Pending	Approve Reject Flag
Figma Designer Intern	Meta	Internship	Pending	Approve Reject Flag
Software Engineer Intern	Google	Internship	Pending	Approve Reject Flag
AI Research Assistant	MIT CSAIL	Research	Pending	Approve Reject Flag
FrontEnd Developer	Shohoz	Job Post	Pending	Approve Reject Flag
DSA Research Assistant	Harvard	Research	Pending	Approve Reject Flag
Figma Designer Intern	Meta	Internship	Pending	Approve Reject Flag

Admin Review posting page

 **NexusPortal**
Research Management

Admin Dashboard

System Overview

Verify Users

Review Postings

Reports

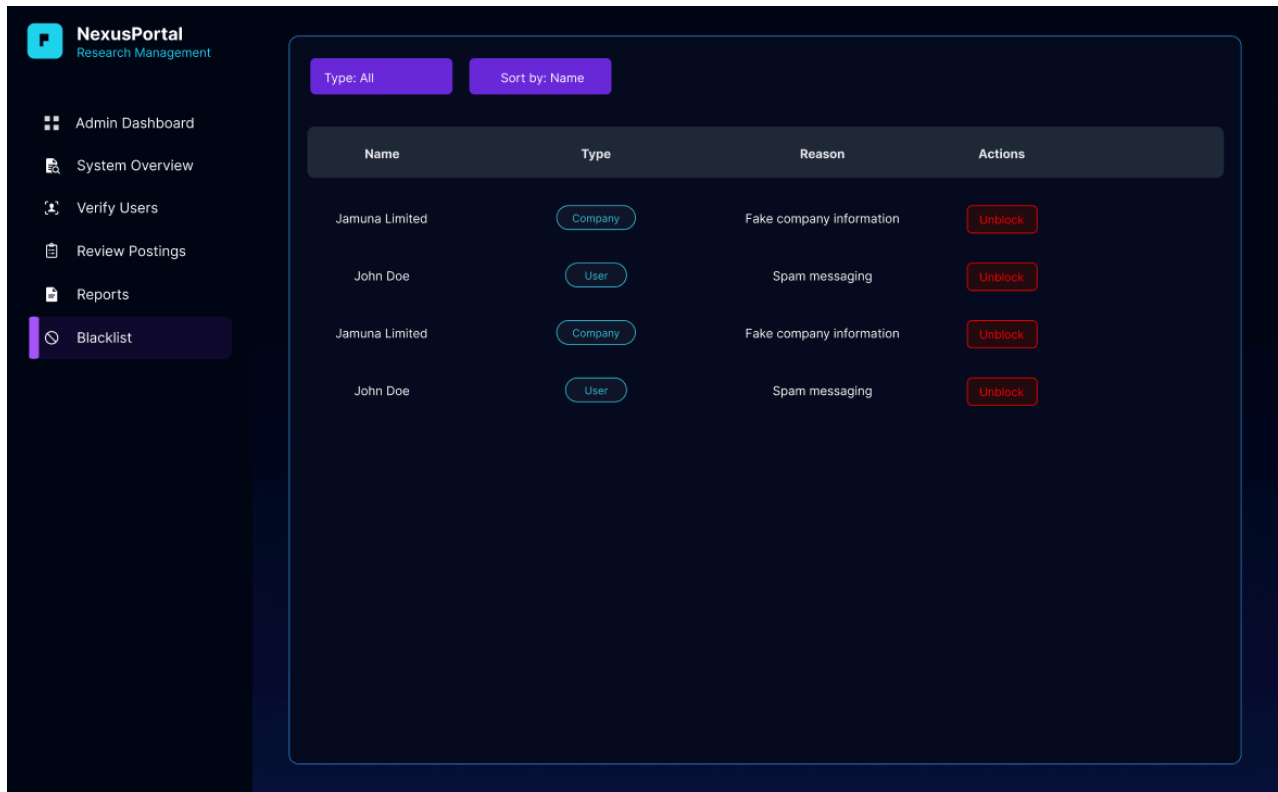
Blacklist

Role: All

Status: Pending

Name	Role	Documents	Status	Actions	
Robin peter	Faculty	Business License	Pending	Approve	Reject
Nazmul Kabir	Employer	Business License	Pending	Approve	Reject
Razin Rahman	Student	Business License	Pending	Approve	Reject
Tushar Rahman	Employer	Business License	Pending	Approve	Reject
Priya	Student	Business License	Pending	Approve	Reject
Robin peter	Faculty	Business License	Pending	Approve	Reject
Nazmul Kabir	Employer	Business License	Pending	Approve	Reject
Razin Rahman	Student	Business License	Pending	Approve	Reject
Tushar Rahman	Employer	Business License	Pending	Approve	Reject
Priya	Student	Business License	Pending	Approve	Reject

Admin Reports page



Admin Blacklist page

NexusPortal

Research Management

Overview Dashboard

Overview of your recruitment activities

Log Out

Employer Dashboard

12

Total Jobs Posted

248

Total Applications Received

34

Shortlisted Candidates

8

Interviews Scheduled

Active Job Posts

Job Title	Job Type	Posted Date	Total Applicants	Status	Action
Senior Developer	Full-time	2025-12-15	45	Active	View Edit Close
Product Manager	Full-time	2025-12-20	32	Active	View Edit Close
UX Designer	Full-time	2025-12-22	28	Active	View Edit Close
Data Analyst	Internship	2026-01-02	56	Active	View Edit Close
Marketing Manager	Full-time	2026-01-03	41	Active	View Edit Close
Research Assistant	RA	2025-11-10	18	Active	View Edit Close

PD

Prof. Davis

Supervisor

Employer Dashboard page

55

NexusPortal
Research Management

Employer Dashboard

Post a Job

My Jobs

Applications

Messages

Settings

PD

Prof. Davis
Supervisor

Job Title

Job Category

Location

Job Description

Required Skills

HTML

CSS

+

Experience Level

Company Name

Job Type

Full-time

Part-time

Remote

Salary Range

+

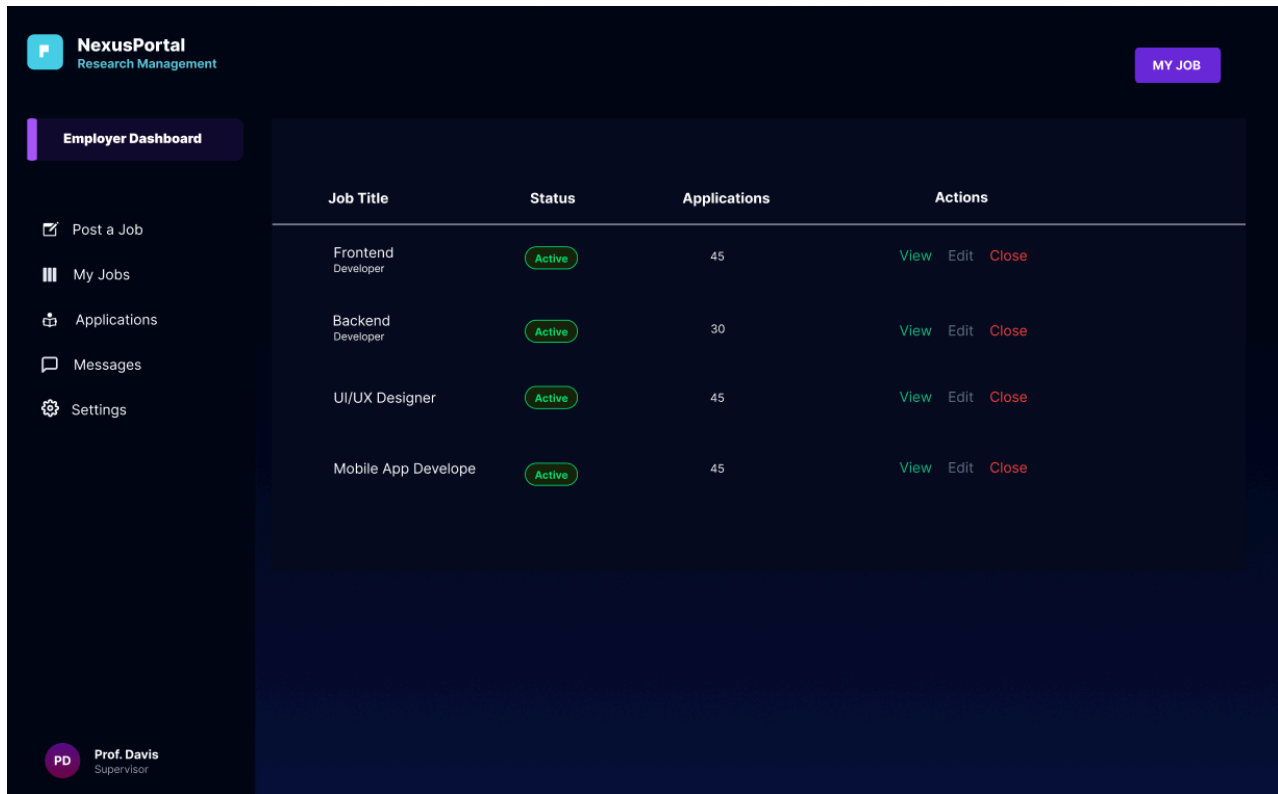
Deadline

Post Job

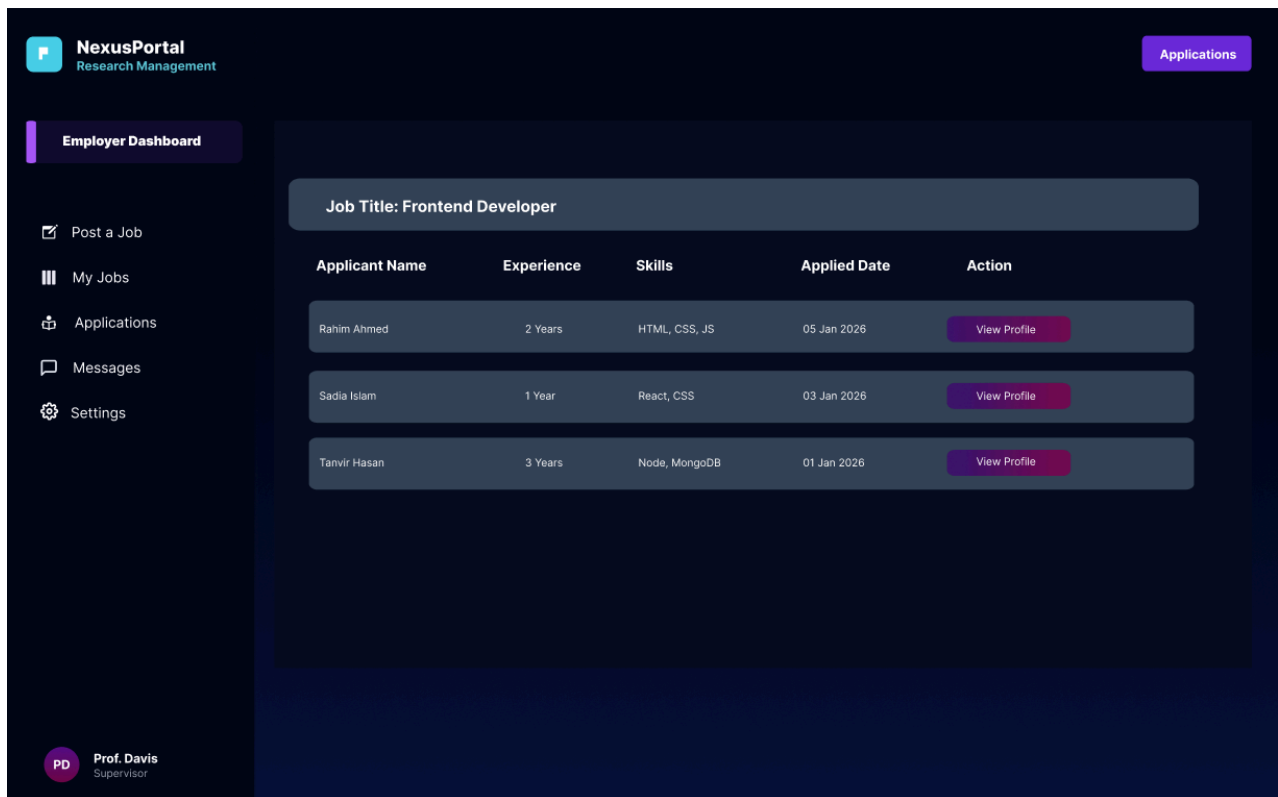
Cancel

Post Job

Employer Post a job page



Employer My Jobs page



Employer Applications Page

NexusPortal

Research Management

Profile & Settings

Employer Dashboard

Post a Job

My Jobs

Applications

Messages

Settings

Company Name

Enter Company Name

Logo Upload

Upload Logo

Contact Info

Email

Phone Number

Change Password

New Password

Confirm Password

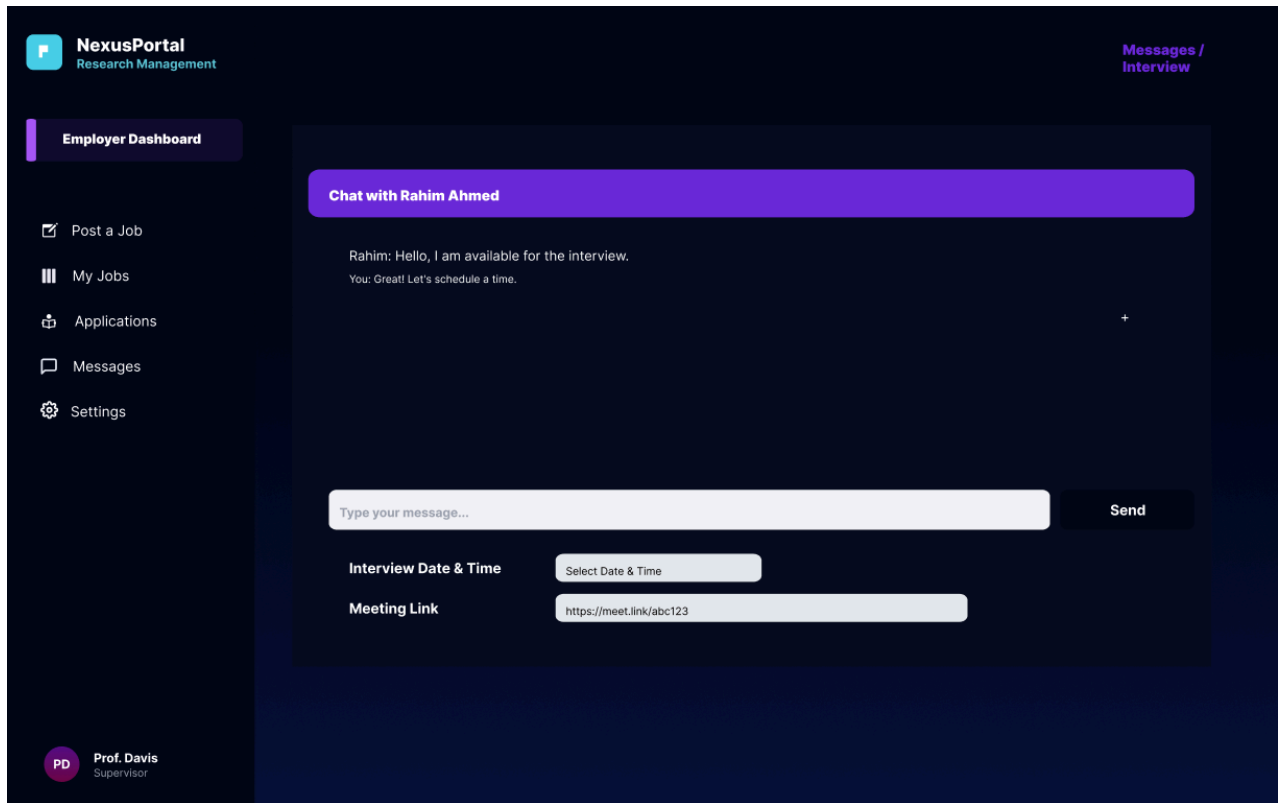
Save Changes

PD

Prof. Davis

Supervisor

Employer settings page



Employer messages page

8.2 UI Design Principles

The user interface of Nexus Portal is designed following standard Human–Computer Interaction (HCI) guidelines, mainly inspired by Shneiderman’s Eight Golden Rules and Don Norman’s design principles.

- Consistency is maintained across all dashboards, buttons, colors, and layouts so users can easily understand and navigate the system.
- Visibility of system status is ensured using status badges, counters, and notifications (e.g., application status, verification status).

- User control and simplicity are prioritized through clear navigation, minimal layouts, and clearly labeled actions.
- Recognition over recall is supported by persistent menus, icons, and dashboards that summarize key information.
- The system follows a responsive and mobile-friendly design, allowing smooth usage across both web and mobile devices.

Overall, the UI focuses on clarity, ease of use, and role-based interaction, ensuring an efficient experience for students, faculty, employers, and administrators.

9. Conclusion

The NexusPortal platform is designed to address the fragmentation of career-related opportunities by integrating jobs, internships, and academic research positions into a single, secure, and verified system. By supporting multiple user roles—students, employers, university faculty, administrators, and reviewers—the platform ensures structured interaction, transparency, and accountability throughout the opportunity lifecycle.

The system emphasizes verified postings, role-based access control, application tracking, and research collaboration, which differentiates it from traditional job portals. Its modular design enables scalability, maintainability, and future enhancement while remaining feasible within academic and resource constraints. The inclusion of features such as resume building, notification management, and analytics further enhances usability and effectiveness for all stakeholders.

This Software Requirements Specification (SRS) document clearly defines the system scope, functionality, operating environment, constraints, and assumptions, serving as a comprehensive reference for system design and implementation. The NexusPortal, when implemented as specified, has the potential to significantly improve opportunity accessibility, recruitment efficiency, and research collaboration within academic and professional environments.

10. References

The following sources were used in the preparation of this Software Requirements Specification:

1. Project Documentation:

- Karim, R., Hossen, M. Z., Emon, A. G., Sameer, M. S. R., & Hasan, J. (2026). *NexusPortal - Feasibility Analysis Report*. United International University .
- Team_3. (2026). *NexusPortal - Unified Career Platform Project Proposal*. United International University.

2. Market Research & Data:

- "NexusPortal Market Gap Survey." Internal Project Survey (n=68 participants), Conducted via Google Forms, December 2025 .
- "Faculty Recruitment Insights." Qualitative discussion with UIU Faculty members regarding Research Assistant recruitment challenges .

3. Technical Documentation:

- React Documentation. *Meta Open Source*. Available at: <https://react.dev/>.
- Node.js Documentation. *OpenJS Foundation*. Available at: <https://nodejs.org/en/docs/>.
- PostgreSQL Documentation. *The PostgreSQL Global Development Group*. Available at: <https://www.postgresql.org/docs/>.
- Tailwind CSS Documentation. Available at: <https://tailwindcss.com/docs>.

4. Academic Textbooks:

- Sommerville, I. (2015). *Software Engineering* (10th ed.). Pearson Education. (Reference for SDLC and SRS structure).
- Pressman, R. S. (2014). *Software Engineering: A Practitioner's Approach* (8th ed.). McGraw-Hill Education.

11. Appendices

Appendix A – Glossary

A

- **API (Application Programming Interface):** A set of functions and procedures allowing the creation of applications that access the features or data of an operating system, application, or other service.
- **ATS (Application Tracking System):** A software application that enables the electronic handling of recruitment and hiring needs. In NexusPortal, this refers to the Kanban-style board used by students.

C

- **Client-Server Architecture:** A distributed application structure that partitions tasks or workloads between the providers of a resource or service, called servers, and service requesters, called clients.

F

- **Frontend:** The graphical user interface of a website or software application that users can view and interact with (developed here using React.js).

K

- **Kanban:** A visual method for managing work as it moves through a process. NexusPortal uses this for tracking application statuses (e.g., Applied $\rightarrow$ Interview $\rightarrow$ Offer).

M

- **MFA (Multi-Factor Authentication):** A security system that requires more than one method of authentication from independent categories of credentials to verify the user's identity.

P

- **PostgreSQL:** An open-source relational database management system (RDBMS) emphasizing extensibility and SQL compliance.
- **Puppeteer:** A Node.js library which provides a high-level API to control Chrome or Chromium; used in NexusPortal for generating PDF resumes.

R

- **RA (Research Assistant):** A student employed by a university or faculty member to assist with academic research tasks.
- **RBAC (Role-Based Access Control):** A method of restricting network access based on the roles of individual users within an enterprise (e.g., distinguishing between Student and Faculty access).

S

- **SAD:** System Analysis and Design.
- **SDLC (Software Development Life Cycle):** A process for planning, creating, testing, and deploying an information system.
- **SRS:** Software Requirements Specification.

U

- **UI/UX:** User Interface / User Experience.
- **Unified Dashboard:** The central feature of NexusPortal that combines Jobs, Internships, and Research opportunities into a single view.

V

- **Verification Badge:** A visual indicator (checkmark) on a user profile confirming that their Student ID or Academic Transcript has been manually validated by an Administrator.